Lecture No -1

Introduction

India's geo-climatic conditions as well as its high degree of socio-economic vulnerability, makes it one of the most disaster prone country in the world. A disaster is an extreme disruption of the functioning of a society that causes widespread human, material, or environmental losses that exceed the ability of the affected society to cope with its own resources. Disasters are sometimes classified according to whether they are "natural" disasters, or "human-made" disasters. For example, disasters caused by floods, droughts, tidal waves and earth tremors are generally considered "natural disasters." Disasters caused by chemical or industrial accidents, environmental pollution, transport accidents and political unrest are classified as "human-made" or "human-induced" disasters since they are the direct result of human action.

A more modern and social understanding of disasters, however, views this distinction as artificial since most disasters result from the action or inaction of people and their social and economic structures. This happens by people living in ways that degrade their environment, developing and over populating urban centres, or creating and perpetuating social and economic systems. Communities and population settled in areas susceptible to the impact of a raging river or the violent tremors of the earth are placed in situations of high vulnerability because of their socio-economic conditions. This is compounded by every aspect of nature being subject to seasonal, annual and sudden fluctuations and also due to the unpredictability of the timing, frequency and magnitude of occurrence of the disasters.

Etymology

The word 'Disaster' derives from Middle French *désastre*[1] and that from Old Italian *disastro*, whichin turn comes from the Greek pejorative prefix *δυσ-*, (*dus-*) "bad"+ □*στήρ* (*aster*), "star". The root ofthe word *disaster*[2] ("bad star" in Greek and Latin) comes from an astrological theme in which theancients used to refer to the destruction or deconstruction of a star as a disaster.

Definition

Disaster is an event or series of events, which gives rise to casualties and damage or loss of properties, infrastructures, environment, essential services or means of livelihood on such a scale which is beyond the normal capacity of the affected community to cope with. Disaster is also sometimes described as a *"catastrophic situation in which the normal pattern of life or eco-system has been disrupted and extra-ordinary emergency interventions are required to save and preserve lives and or the environment".*

The Disaster Management Act, 2005 defines disaster as *"a catastrophe, mishap, calamity or grave occurrence in any area, arising from natural or man made causes, or by accident or negligencewhich results in substantial loss of life or human suffering or damage to, and destruction of, property,or damage to, or degradation of, environment, and is of such a nature or magnitude as to be beyondthe coping capacity of the community of the affected area".*

The United Nations defines disaster as *"the occurrence of sudden or major misfortune which disrupts the basic fabric and normal functioning of the society or community"*

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Box 1.1: List of various Disasters	
i. Water and climate relateddisasters	a) Floods and drainage management b) Cyclones c) Tornadoes and hurricanes d) Hailstorm e) Cloud burst f))Heat wave and cold wave g) Snow avalanches h) Droughts i) Sea erosion j) Thunder and lightening k) Tsunami
ii. Geological related disasters	a) Landslides and mudflows b) Earthquakes c) Dam failures/ Dam bursts d) Minor fires
iii. Chemical, industrial and nuclear industrial disastersrelated disasters	a) Chemical and b) Nuclear disasters
iv. Accident related disasters	a) Forest fires b) Urban fires c) Mine flooding d) Oil spills e) Major building collapsef) Serial bomb blasts g) Festival related disasters h) Electrical disasters and fires i) Air, road and rail accidents j) Boat capsizing k) Village fire
v. Biological related disasters	a) Biological disasters and epidemics b) Pest attacks c) Cattle epidemics d) Food poisoning

HAZARD AND DISASTERS

Hazard is a danger defined as a phenomenon or natural condition having the potential of causing loss of lives, injury and destruction of properties. Overall it has the possibility of economic disruptions and environmental damage.
hazard is called a disaster only when it really affects a certain population. The occurrence of natural disasters cannot be controlled by human beings but with better mitigation and management measures their impacts may be minimised.

Due to human intervention in the natural processes, the destructive magnitude and frequencyof natural disasters have increased considerably in recent times. The severity of disastersdepends on our preparedness. The scale of the impact in turn depends on the choices we dohave. The choices are concerned with our capabilities to cope with the disaster. For examplea well built house has a higher possibility of coping with the wrath of nature than ill-built houses. It is called vulnerability. Well built houses are less vulnerable while the ill-built housesare highly vulnerable to

disasters.

Hence, disaster is defined as: "… a serious disruption of the functioning of a society, causingwidespread human, material, or environmental losses which exceed the ability of the affectedsociety to cope using its own resources." The United Nations defines disaster as "…theoccurrence of a sudden or major misfortune which disrupts the basic fabric and normalfunctioning of a society or community."

It is an event or a series of events which gives rise to casualties and/or loss of property,infrastructure, essential services or means of livelihood on a scale that is beyond the normalcapacity of the affected communities to cope with. Disaster is sometimes also used to describea "catastrophic situation in which the normal patterns of life or ecosystems have been disruptedand extraordinary emergency interventions are required to save and preserve human lives and / or the environment".

a) According to an estimate, about 60,000 people per year are killed by natural disasters and are responsible for 0.1% of global deaths in last 100 years.

b) According to UN statistics, natural disasters cause property damage of around Rs 20,000 crores worldwide every year.

A disaster occurs when hazard impacts vulnerable people or communities of a certain area. Ithappens when their inability to reduce the potential negative consequences surpasses theircapacity to cope.

(Natural Hazard + Vulnerability Risk)/ Capacity = DISASTER

DISASTER RISK

Risk is the potential disaster which causes losses of lives, properties and infrastructures. It is dependent upon the situation of danger (hazard). Physical, socioeconomic and environmental coping capacity (vulnerability) and the quality of structures the community has exposure to. Hence a common overlapping of these three is called risk.

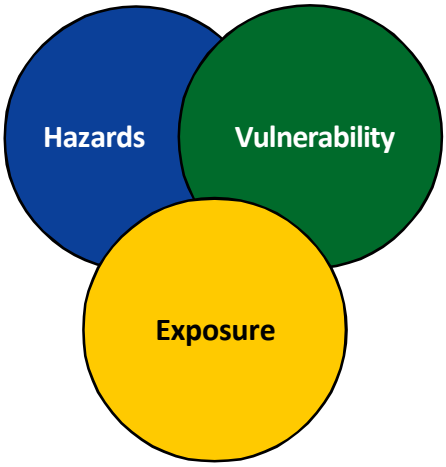


Fig. Identification of Risk

A single disaster has a different impact on different people in the society. Poor people have less capacity to cope with the disaster in comparison to rich people. Because poor people have less strong houses constructed in low economic zones, situated in low lying areas with

less economic power. All are making them more vulnerable to any disaster. They are more exposed to nature and subject to risk. On the other hand, the reverse is the case with rich people. Therefore risk is determined by hazard, vulnerability and exposure of the people to the disaster.

The Characteristic of hazard and disaster are summarised in Table

Natural Hazards and Disasters

Table Differences between Natural Hazard and Disaster

	Natural Hazard	Disaster
1.	Hazards are dangerous physical conditions or events.	1. Most of the disasters occur rapidly, instantaneously and indiscriminately.
2.	Hazards have potential of damaging different forms of lives	2. Disasters are largely viewed from a human perspective causing severe damage to human life and property.
3.	Hazard represents a latent threat to damage biotic and abiotic components of the environment.	3. Disaster disrupts the normal functioning of society and the physical environment.
4.	Hazards may or may not turn into disasters.	4. All disasters cause damage to property and loss of lives. A large number of people are affected.
5.	External aid is not required to damage biotic and abiotic components of the environment.	5. It affects the society and socio, economic and physical environment to such an extent that external aid becomes necessary.
6.	Earthquakes, floods, cyclones, volcanic eruptions, landslides, droughts etc are called natural hazards before they cause loss of life and damage to property.	6. Earthquakes, floods, cyclones, volcanic eruptions, landslides, droughts etc are called natural disasters after they cause loss of life and damage to property.
7.	People are not affected.	7. People are affected
8.	Hazards also occur in areas not	8. Hazards turn into disasters when they

occupied by human beings.	occur in the inhabited areas with infrastructures, buildings, telecommunications etc.
9. Hazards are processes of the genesis of extreme events	9. Disasters are the responses to the aftermath of natural hazards.

TYPES OF NATURAL DISASTERS

Natural disasters are those which are caused by nature and are beyond human control. But itis also true that rampant human activities accelerate the impact of natural disasters. For examplea judicious use of mountain slope, over irrigation. Deforestation, road construction on slopesintensifies the impact of natural disaster landslides. There are several types of natural disasterslike floods, cyclones, earthquakes and volcanic eruptions etc. the types of natural disaster arevery clearly shown in figure.

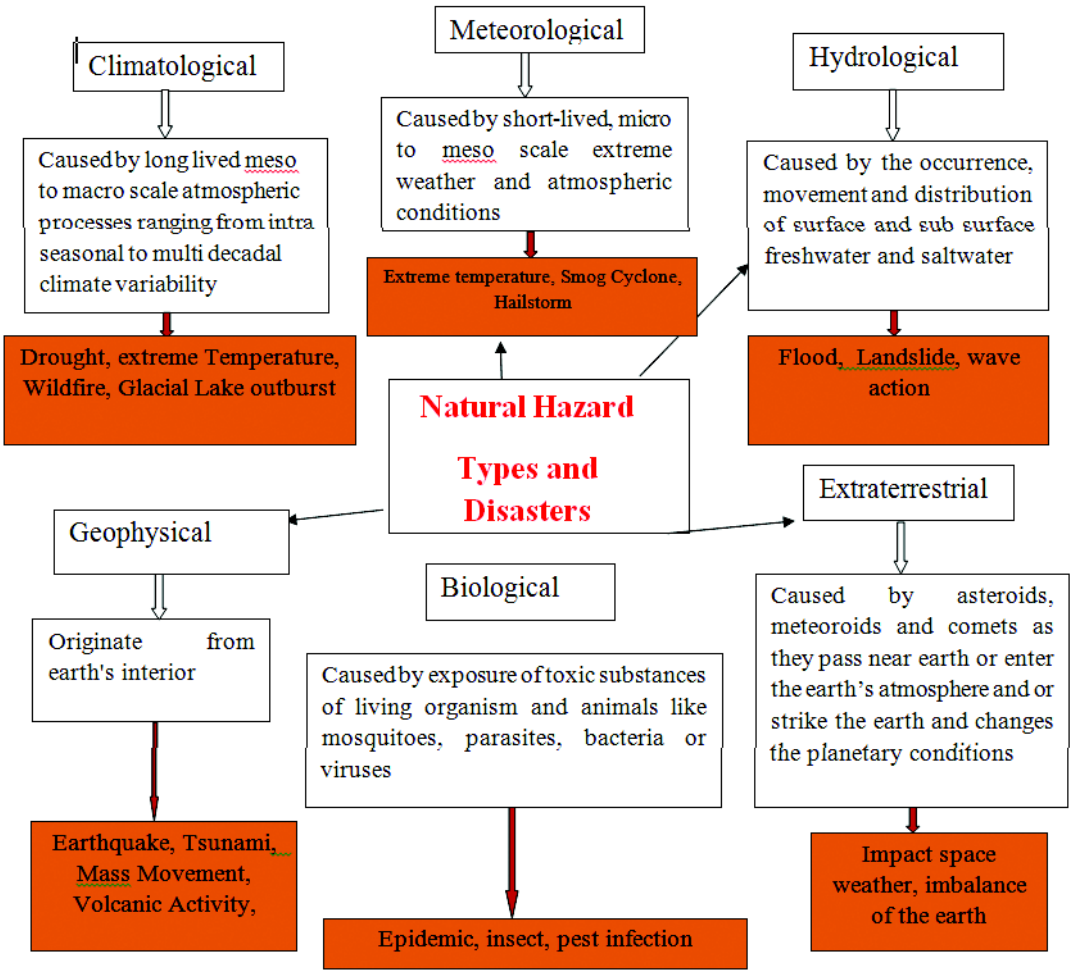


Fig. 13.1 Types of Natural Disaster

CONCEPTS OF DISASTER MANAGEMENT

Disaster management includes various activities for avoiding or reducing ill effects of disaster. It also helps in providing better recovery, bringing normally after disaster as well as preparing

for disaster too. All government, non-governmental and community-based organisations play vital roles in the process. It now includes pre-disaster planning and preparedness activities, organisational planning, training, information management and public relations. Disaster management includes precaution, mitigation, preparedness, response, recovery and reconstruction.

Pre-disaster	**During -disaster**	**Post-disaster**
Preparedness	Response	Response
Prevention	Relief	Recovery
Mitigation		Reconstruction

1. **Disaster Preparedness :** It is a set of measures adopted well in advance for response and to cope with disaster in a better way. Its prime concern is to reduce the loss of human life and property.

2. **Disaster Prevention :** Avoidance of ill impact of disaster is termed as prevention of disaster. It is quite obvious that complete avoidance of any disaster is not possible but it could be minimised. Its compounding effects could be avoided/prevented by building the capacity of individual/society or community.

3. **Disaster Mitigation :** It is the reduction of harmful effects of disasters. It refers to ongoing efforts at different levels starting from individual, local community, to state and national. It includes hazard and disaster mapping and zonation building specific plans for disasters in certain areas prone to it execution of the plan in needed hours is very important for mitigation.

4. **Disaster Relief :** It is immediate help and support made available to affected people. It is of great importance to minimise suffering and extend basic minimum needs to support life. It includes providing food, water, shelter and medical care.

5. **Disaster Response :** It refers to the action taken up immediately after the disaster happened. This includes many efforts like rescue, establishment of marginal health care, treating injured supplying eatables, shifting people to safe places etc. the first and immediate response is generally provided by local youths, volunteers and later on, the action of district, state or local level help arrives.

6. **Recovery after Disaster:** It refers to providing help to affected people in a shortest possible time. It depends upon the ability of the organisation/efforts taken by the NGOs district state or national level disaster management authority

7. **Disaster Reconstruction :** it refers to the construction of a situation again after its destabilisation/imbalance due to disaster. It involves partial or complete relocation and reestablishment of essential infrastructures and shelter. It helps in reducing the trauma to some extent, especially related to physical facilities needed for infected people.

Lecture No-2

Volcano

A volcano is an opening in the earth's crust through which magma, gases and ash are released to the earth's surface. The molten rock material found in the interior of the earth is called magma. It can be noted that when magma reaches the earth's surface, it is known as lava (Figure. 3.25). Vent is an opening or mouth of a volcano. Fumaroles are the gushing fumes through the gap in the volcano. Crater is a saucer shaped depression in the mouth of a volcano. When the crater is widened, it is called as Earthquake,Iran-Iraq border, 2017 Caldera. Volcanic ash consists of fragments of pulverized rock, minerals and volcanic glass, created during volcanic eruptions. Volcano generally erupts either through the vent (E.g. Mt. Fujiyama, Japan) or fissure (The Deccan Plateau, India).

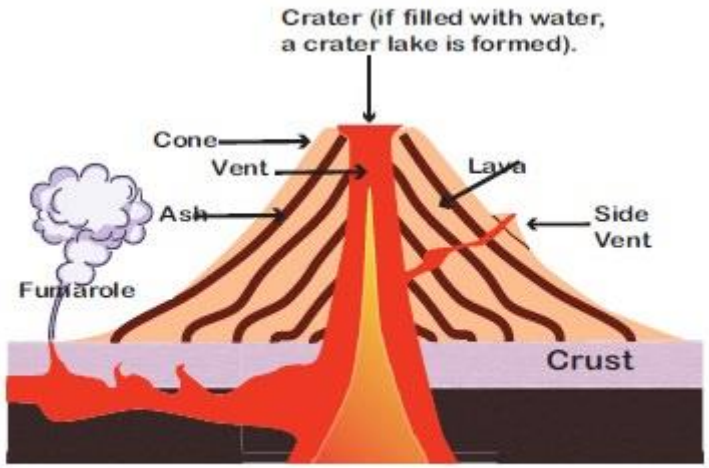


Figure 3.25 Volcano

Pumice is a volcanic rock produced when lava with a very high content of water and gases is discharged from a volcano.

Causes of Volcanic Eruptions

The following are the causes of volcanic eruptions:

Weak Zones in the Earth Crust: The parts of the earth where two tectonic plates collide against or drift apart from each other are considered very weak. Volcanoes may erupt in such zones, for example, African and Eurasian plates.

Magma Saturated with Gases: The magma, in the interior of the earth, is often found saturated with gases like carbon dioxide, and hydrogen sulfide. These gases together with water vapour make the magma highly explosive. Magma is forced out as lava on the surface of the earth due to the pressure exerted by these gases.

Types of Volcanoes

Based on the **frequency of eruption**, there are three types of volcanoes:

1. Active Volcanoes: Volcanoes which erupt frequently are called active volcanoes. Generally, their vent remains open. Mount Etna of Italy, Cotopaxi in Ecuador are some examples.

2. Dormant Volcanoes: These volcanoes may not have erupted in the recent past but there is a possibility of eruption at any time. In other words, they may lie dormant awaiting active eruption anytime. Sometimes gases and steam come out of them. They cause great destruction to life and property once they become active again. Mt. Vesuvius of Italy and Mt. Fujiyama of Japan are examples.

3. Extinct Volcanoes: These volcanoes have exhausted their energy and have not erupted during the known geological period. The vent of these volcanoes remains closed with solidified lava. The formations such as craters may be filled with water and crater lakes may be formed. The slopes of these landforms may be covered with vegetation. Popa in Myanmar and Mt. Kenya in eastern Africa are the examples of extinct volcano.

On the **basis of nature of eruption and form** developed on the surface, they are classified into following types:

1. Shield Volcanoes: These are made up of basalt, a type of lava that is very fluid when erupted. They become explosive when water gets into the vent. They develop into a cinder cone. Hawaiian volcano is an example of this category.

2. Composite cone volcanoes: They are also called 'strato volcanoes'. They are cone-shaped volcanoes composed of layers of lava, ash and rock debris. Mount Vesuvius and Mount St. Helens are examples of composite volcanoes.

3. Cinder Cone Volcano : It forms when magma is thrown out to the surface, cooled in to ash and cinders and settled around the mouth of volcano. It is less dangerous than other volcanoes.

4. Lava Dome: Unlike composite and shield volcanoes, lava domes are of significantly smaller structure. They are formed when the lava is too viscous to flow to a great distance. As the lava dome slowly grows, the outer surface cools and hardens as the lava continues to pile within. Eventually, the internal pressure can shatter the outer surface, causing loose fragments to spill down its sides.

Fact File

The greatest volcanic explosion known to humans is perhaps Mt. Krakatau in August 1883.

Krakatau is a small volcanic island in the Sunda Straits, between Java and Sumatra.

The explosion could be heard in Australia, almost 4,000 km away.

The vibration set up enormous waves over 30 m high which drowned 36,000 people in the coastal districts of Indonesia.

Effects of Volcanic Activities

Showers of cinders and bombs can cause damage to life and properties. Sometimes ash can precipitate under the influence of rain and completely cover large areas.

The volcanic gases pose potential hazard to people, animals; agriculture, while sulfur dioxide gas can lead to acid rain and air pollution.

Positive Effects of Volcanoes

Volcanism creates new landforms. Volcanic rocks yield very fertile soil upon weathering and decomposition.

The Kimberlite rock of South Africa, the source of diamonds, is the pipe of an ancient volcano.

In the vicinity of active volcanoes, waters in the depth are heated from contact with hot magma giving rise to springs and geysers. The Puga valley in Ladakh region and Manikaran (Himachal Pradesh) are promising spots in India for the generation of geothermal electricity.

Distribution of Volcanoes across the World

Most known volcanic activity and the earthquakes occur along converging plate margins and mid-oceanic ridges. The major regions of volcanic distributions are as follows.

1. Pacific Ring of Fire

Circum-Pacific region, popularly termed the 'Pacific Ring of Fire', has the greatest concentration of active volcanoes. Volcanic belt and earthquake belt closely overlap along the 'Pacific Ring of Fire'. It is estimated to include two-thirds of the world's volcanoes.

2. Mid Atlantic Region

The Mid Atlantic Region coasts has comparatively fewer active volcanoes but many dormant or extinct volcanoes, example. St. Helena, Cape Verde Islands and the Canary Islands. But the volcanoes of Iceland and the Azores are active.

3. The Great Rift valley of Africa

In Africa some volcanoes are found along the East African Rift Valley. Kilimanjaro and Mt. Kenya are extinct volcanoes. The only active volcano in West Africa is Mt. Cameroon.

4. Mediterranean Region

Volcanoes of the Mediterranean region are mainly associated with the Alpine folds. Example, Mt. Vesuvius, Mt. Stromboli (known as the Light House of the Mediterranean Sea).

5. Other Regions

Elsewhere in the interiors of continents of Asia, North America and Europe active volcanoes are rare. There are no volcanoes in Australia.

Volcanoes in India

There are no volcanoes in the Himalayan region of India. However, Barren Island, lying 135 km north-east of Port Blair became active in 1991 and 1995.

However, the other volcanic island in Indian Territory is Narcondam (Andaman and Nicobar Islands) It is probably extinct. Its crater wall has been completely destroyed.

Volcanic eruptions can have many hazardous effects, including:

- Health hazards

  Volcanic eruptions can cause short-term health issues like eye and airway irritation, vomiting, dizziness, and headaches. Long-term exposure can lead to more serious conditions like bronchitis, lung infection, and lung disease.

- Ash fall

  Ash can travel hundreds or thousands of miles downwind from a volcano. It can be abrasive, gritty, and sometimes corrosive. Ash can be especially dangerous for infants, the elderly, and people with respiratory issues. Ash can also damage or clog drinking water and wastewater treatment facilities.

- Volcanic gases

  Volcanic gases like sulfur dioxide, carbon dioxide, and hydrogen fluoride can be harmful to health, vegetation, and infrastructure. Sulfur dioxide can cause acid rain and air pollution.

- Ballistic projectiles

  Volcanoes can hurl rocks into the air that travel like cannonballs. These rocks can land within 2 km of the vent, but can travel much further during explosive eruptions.

- Tsunamis

  Underwater volcanic eruptions can cause tsunamis, which are seismic sea waves.

- Climate change
  Volcanic eruptions can impact the climate in a number of ways:
  - Cooling effect: Sulfur dioxide emitted during eruptions can reflect sunlight and cool the Earth's climate.
  - Ozone depletion: Sulfur dioxide can also contribute to ozone depletion.
  - Greenhouse effect: Volcanic eruptions can release carbon dioxide and other halogens into the stratosphere, which can cause a greenhouse effect.

Volcanic eruptions can have many effects on the environment, including:

- Climate change: Volcanoes release large amounts of gases into the atmosphere, including sulfur dioxide and carbon dioxide. Sulfur dioxide can cause global cooling, while carbon dioxide is a greenhouse gas that can promote global warming.
- Air quality: Volcanic eruptions create smog and harmful gases that can irritate the skin, eyes, nose, and throat, and make it difficult to breathe.
- Water contamination: Volcanic eruptions can contaminate water supplies.

- Secondary events: Volcanic eruptions can cause secondary events like floods, landslides, and mudslides if there is rain, snow, or melting ice.
- Wildfires: Hot ashes from volcanic eruptions can start wildfires.
- Health threats: Volcanic eruptions can pose health threats, including suffocation, infectious diseases, respiratory diseases, burns, and traumatic injuries.
- Visibility reduction: Volcanic eruptions can reduce visibility.
- Machinery damage: Volcanic eruptions can damage machinery.
- Loss of basic amenities: Volcanic eruptions can damage power lines, gas pipelines, water and sewage pipes, leading to the loss of basic amenities like fresh water, gas, and electricity.

Lecture 3

Earthquake Hazards

Misconceptions

Earthquakes really pose little direct danger to a person. People can't be shaken to death by an earthquake. Some movies show scenes with the ground suddenly opening up and people falling into fiery pits, but this just doesn't happen in real life.

The Effect of Ground Shaking

The first main earthquake hazard (danger) is the **effect of ground shaking**. Buildings can be damaged by the shaking itself or by the ground beneath them settling to a different level than it was before the earthquake (**subsidence**).

Buildings can even sink into the ground if soil liquefaction occurs. **Liquefaction** is the destabilization of sand or soil when **groundwater** (water underground) is forced out from between grains during the shaking of a moderate or strong earthquake. If liquefaction occurs under a building, it may start to lean, tip over, or sink several feet. The ground firms up again after the shaking has stopped. Liquefaction is a hazard in areas that have groundwater near the surface and sandy soil.

Buildings can also be damaged by strong surface waves making the ground heave and lurch. Any buildings in the path of these surface waves can lean or tip over from all the movement. The ground shaking may also cause landslides, mudslides, and avalanches on steeper hills or mountains, all of which can damage buildings and hurt people.

Ground Displacement

The second main earthquake hazard is **ground displacement** (ground movement) along a fault. If a structure (a building, road, etc.) is built across a fault, the ground displacement during an earthquake could seriously damage or rip apart that structure.

From the figure on the left you can tell that the San Andreas Fault is a **right-lateral** transverse (strike-slip) fault because the other side of the road (on the opposite side of the fault) has moved to the right, relative to the photographer's position.

Flooding

The third main hazard is **flooding**. An earthquake can **rupture** (break) dams or levees along a river. The water from the river or the reservoir would then flood the area, damaging buildings and maybe sweeping away or drowning people.

Tsunamis and seiches can also cause a great deal of damage. A **tsunami** is what most people call a tidal wave, but it has nothing to do with the tides on the ocean. It is a huge wave caused by an earthquake under the ocean. Tsunamis can be tens of feet high when they hit the shore and can do enormous damage to the coastline. Seiches are like small tsunamis. They occur on lakes that are shaken by the earthquake and are usually only a few feet high, but they can still flood or knock down houses, and tip over trees.

Fire

The fourth main earthquake hazard is **fire**. These fires can be started by broken gas lines and power lines, or tipped over wood or coal stoves. They can be a serious problem, especially if the water lines that feed the fire hydrants are broken, too. For example, after the Great San Francisco Earthquake in 1906, the city burned for three days. Most of the city was destroyed and 250,000 people were left homeless.

Most of the hazards to people come from man-made structures themselves and the shaking they receive from the earthquake. The real dangers to people are being crushed in a collapsing building, drowning in a flood caused by a broken dam or levee, getting buried under a landslide, or being burned in a fire.

Earthquakes can cause a variety of hazards, including:

- Ground shaking: Can damage buildings and infrastructure
- Landslides and rockfalls: Can occur due to earthquakes

- Tsunamis: Huge waves caused by an earthquake under the ocean
- Flooding: Earthquakes can rupture dams or levees, causing flooding
- Fires: Can be caused by broken gas lines and power lines, or tipped over wood or coal stoves
- Liquefaction: Can occur due to earthquakes
- Avalanches: Can occur due to earthquakes
- Fissures: Can occur due to earthquakes

The extent of destruction and harm caused by an earthquake depends on several factors, including:

- The magnitude, intensity, and duration of the earthquake
- The local geology
- The time of day that it occurs
- The design and materials of buildings and industrial plants
- The risk-management measures put in place

To reduce the impacts of earthquakes, you can:

- Replace heavy roofing materials with lighter weight options
- Ensure any heavy tiles are properly secured to roof framing beneath
- Make sure your home is securely anchored to its foundations
- Avoid hazardous areas and steer clear of bookcases, heavy furniture, and appliances that may topple over

Earthquakes occur when tectonic plates move suddenly along faults in the Earth's crust, releasing stored-up energy:

Here are some details about earthquakes:

- Tectonic plates

  The Earth's outermost layer is made up of about 15 tectonic plates that move slowly relative to each other.

- Faults

  The edges of tectonic plates meet at faults, where the plates can scrape, bump, or drag against each other.

- Stress and friction

  Tectonic plates are always moving, but friction at their edges can cause them to get stuck. When the stress on the edge overcomes the friction, an earthquake occurs.

- Seismic waves

  The sudden release of energy during an earthquake causes seismic waves that travel through the Earth's crust, causing the ground to shake.

- Hypocenter and epicenter

The location underground where the rock first breaks is called the hypocenter, and the location directly above it on the surface of the Earth is called the epicenter.

- Earthquakes and the Ring of Fire
  The Ring of Fire, a U-shaped belt around the Pacific Ocean, is a hot spot for earthquakes because it's a connection point for many tectonic plates

Earthquakes occur in distinct belts around the world, primarily along plate boundaries:

- The Circum-Pacific Belt

  Also known as the "Ring of Fire", this belt is located around the Pacific Ocean and is the most seismically active zone on Earth. It's where the Pacific plate is subducted beneath the surrounding plates, causing earthquakes in subduction zones. This belt accounts for about 81% of the world's largest earthquakes.

- The Alpide Belt

  This belt extends from Java to Sumatra, through the Himalayas, the Mediterranean, and into the Atlantic. It accounts for about 17% of the world's largest earthquakes.

- The Mid-Atlantic Ridge

  This ridge is located on the floor of the Atlantic Ocean, where two tectonic plates are spreading apart. Most of the mid-Atlantic Ridge is deep underwater.

- The Rift Valleys of East Africa
  This region is also a significant zone of seismic activity.

Earthquakes occur at faults in the Earth, which can be observed at the surface or hidden deep underground

Earthquakes can cause many hazardous effects, including:

- Ground shaking: Can damage buildings, lifelines, and the ground beneath them
- Ground displacement: Can seriously damage or rip apart structures
- Landslides: The shaking causes regions of the rock and soil to slide downhill
- Tsunamis: Huge waves caused by an earthquake under the ocean that can cause destruction both locally and at very distant locations
- Flooding: Earthquakes can rupture dams or levees along a river, causing flooding
- Soil liquefaction: A type of ground failure that can occur during an earthquake
- Fissures: Can occur during an earthquake
- Avalanches: Can occur during an earthquake
- Fires: Can occur during an earthquake

An earthquake is the shaking of the earth. It happens naturally when energy is released, creating waves that travel in all directions. When an earthquake occurs the Earth vibrates and these vibrations are detected by seismographs. Moderate earthquakes happen every day, but big, damaging earthquakes are rare. Earthquakes are more common around the edges of tectonic plates. In India, more earthquakes occur where the Indian Plate meets the Eurasian Plate.

Some terminologies related to Earthquakes are:
Hypocenter: The location below the earth's surface where earthquake starts is called Hypocenter
Epicenter: The Location directly above it on the surface of the earth is called epicenter.
Ritcher scale: Measures magnitude of Earthquake
Mercalli scale: Measures the Earthquake intensity

Earthquakes in India
Earthquakes in India mainly happen due to the Indian tectonic plate is colliding into the Eurasian plate. The peninsular region of India is usually stable but sometimes earthquakes occur on the edges of smaller plates. For example the 1967 Koyna earthquake and the 1993 Latur earthquake happened in these areas.

India is divided into four seismic zones (II, III, IV, V) based on the level of seismicity:

Zone II: Low seismicity
Zone III: Moderate seismicity
Zone IV: High seismicity
Zone V: Very high seismicity (includes areas like the Himalayan region, northeastern states, Kutch, and Andaman & Nicobar Islands)
Zones V and IV cover the entire Himalayan region, North-East India, Western and Northern Punjab, Haryana, Uttar Pradesh, Delhi, and parts of Gujarat.

Most of the peninsular region is in a low-risk zone,

while the northern lowlands and western coastal areas are in a moderate hazard zone.

Types of Indian Earthquakes
In India, earthquakes can be categorized based on their origins and the tectonic settings. Here are the main types:

Type of Earthquake	Description	Example/Region
Tectonic Earthquakes	Caused by the movement of Earth's tectonic plates.	
– Interplate Earthquakes	Occur at the boundaries between two tectonic plates.	Himalayan region
– Intraplate Earthquakes	Occur within a tectonic plate.	Latur earthquake (1993), Maharashtra
Volcanic Earthquakes	Associated with volcanic activity due to the movement of magma beneath the Earth's surface.	Andaman and Nicobar Islands
Induced Earthquakes	Caused by human activities such as mining, reservoir filling, geothermal energy extraction, and oil extraction.	
– Reservoir-Induced	Earthquakes induced by the filling of large dams.	Koyna earthquake (1967), Maharashtra
Collapse Earthquakes	Caused by the collapse of underground caverns or mines. Usually localized and of low magnitude.	Mining areas
Explosion Earthquakes	Caused by explosions, such as nuclear tests or large chemical explosions.	Nuclear tests in Pokhran, Rajasthan

1. Tectonic Earthquakes
These are the most common types of earthquakes in India, caused by the movement of the Earth's tectonic plates. They can be divided into:

Interplate Earthquakes: Happen at the boundaries between two tectonic plates. For example, the Himalayan region has interplate earthquakes because the Indian and Eurasian plates collide there.
Intraplate Earthquakes: Happen within a tectonic plate not at its boundary. The 1993 Latur earthquake in Maharashtra is an example.

2. Volcanic Earthquakes
A special type of earthquake called a volcanic earthquake happens only in areas with active volcanoes. These earthquakes occur when molten rock (magma) is injected into or withdrawn from solid rock causing stress changes. This can make the ground sink or crack. These earthquakes can also happen when rock moves to fill spaces left by magma. Volcanic earthquakes do not mean the volcano will erupt they can happen at any time.

3. Induced Earthquakes
These are caused by human activities such as mining, reservoir-induced seismicity (due to the filling of large dams), geothermal energy extraction, and oil extraction. Examples include:

Reservoir-Induced Seismicity: The Koyna earthquake (1967) in Maharashtra is believed to have been induced by the filling of the Koyna dam reservoir.
4. Collapse Earthquakes
These happen when underground caves or mines collapse. They are usually small and only affect a local area.

5. Explosion Earthquakes
These are caused by explosions, such as nuclear tests or large chemical explosions. For example, nuclear tests conducted in Pokhran, Rajasthan, generated minor seismic activity.

Earthquake Zones in India
Complete List of All Zones of Earthquakes in India:

The zones are identified using the Modified Mercalli (MM) intensity, which measures how earthquakes affect areas. After the Killari earthquake in Maharashtra in 1993 the seismic map was updated. The low danger zone called Seismic Zone I was combined with Seismic Zone II. So Zone I is no longer shown on the map.

Zone II
It is a low-intensity area, covering 40.93% of the country's land. This includes the Karnataka Plateau and the peninsula region.

Zone III

This region has moderate intensity and covers 30.79% of the country's area. It includes Kerala, Goa, and the Lakshadweep Islands, along with parts of Punjab, Rajasthan, Madhya Pradesh, Bihar, Jharkhand, Chhattisgarh, Maharashtra, Odisha, and Tamil Nadu.

Zone IV

This is called a high-intensity zone. It covers 17.49% of the country's land. It includes the rest of Jammu & Kashmir, Himachal Pradesh, Delhi, Sikkim, northern parts of Uttar Pradesh, Bihar, West Bengal, the western coast of Maharashtra, and Rajasthan.

Zone V

This is an extremely severe zone. It covers 10.79% of the country's land. It includes North Bihar, Himachal Pradesh, Uttarakhand, the Rann of Kutch in Gujarat, and the Andaman and Nicobar Islands.

Major Earthquakes in India List

Some of the devastating earthquakes have affected India. More than 58.6% of Indian Territory is vulnerable to earthquakes of moderate to very high intensity.

India's most significant earthquakes include:

Kutch Earthquake (1819) which was 8.3 magnitude
Assam Earthquake (1897)
Bihar-Nepal Earthquake (1934) of 8.4 magnitude
Koyna Earthquake (1967) of 6.5 magnitude
Uttarkashi (1991) of 6.6 magnitude
Killari (1993) of 6.4 magnitude
Bhuj (2001) of 7.7 magnitude
Jammu Kashmir (2005)

Key Regions Prone to Different Types of Earthquakes

Region Type of Earthquake Description

Himalayan Region Tectonic (Interplate) Frequent earthquakes due to the collision of the Indian and Eurasian plates.

Northeastern India Tectonic (Interplate) Complex tectonic interactions cause frequent earthquakes.

Western India (Kutch) Tectonic (Intraplate) Earthquake (2001).

Peninsular India Tectonic (Intraplate) Generally stable but can experience intraplate earthquakes like Latur (1993).

Andaman & Nicobar Islands Tectonic & Volcanic Prone to both tectonic and volcanic earthquakes.

List of Major Earthquakes in India Year-wise for UPSC

Year-wise Major Earthquakes

2015 India/Nepal Earthquake
2011 Sikkim Earthquake
2005 Kashmir Earthquake
2004 Indian Ocean Earthquake
2001 Bhuj Earthquake

1999 Chamoli Earthquake
1997 Jabalpur Earthquake
1993 Latur Earthquake
1991 Uttarkashi Earthquake
1941 Andaman Islands Earthquake
1975 Kinnaur Earthquake
1967 Koynanagar Earthquake
1956 Anjar Earthquake
1934 Bihar/Nepal Earthquake
1905 Kangra Earthquake

Causes of Earthquakes in India

Avalanches and Landslides

Tremors can make slopes unstable and collapse causing debris to fall and create landslides mainly in hilly areas. Earthquakes can also trigger avalanches, making large amounts of ice fall from snowy peaks. For example, the 2015 Nepal earthquake caused several avalanches on and near Mount Everest.

The 2011 Sikkim earthquake caused landslides and significant property damage, especially at the Singik and Upper Teesta hydroelectric projects.

Floods

Earthquakes can cause flash floods and make dams and reservoirs fail. Avalanches and landslides can block rivers, leading to flooding. For example, the 1950 Assam earthquake created a barrier in the Dihang River with huge debris, causing flash floods upstream.

Tsunami

Tsunamis are waves created when an ocean basin is disturbed displacing a large amount of water. Earthquakes can move the seafloor, causing these big waves. On December 26, 2004, an earthquake off Sumatra's coast caused the Indian Ocean Tsunami.

This happened because the Indian plate moved under the Burmese plate. Over 240000 people died in the Indian Ocean region and nearby countries. In 2011 a massive undersea earthquake in Japan caused 10-meter tsunami waves during the Tohoku earthquake. This led to a nuclear meltdown at Fukushima Daiichi, causing major global concerns due to radioactive fallout.

Impact of Earthquakes in India

Loss of Human Life and Property

Human towns and buildings suffer severe damage when the ground moves up, down, or sideways. For example, the 2015 Nepal earthquake caused massive urban destruction. This 7.8-magnitude earthquake was 8.2 kilometers deep. Many lives were lost due to uncontrolled urban growth, poorly built structures, and unscientific designs. Kathmandu's urban areas were badly hit causing 8000 deaths and $10 billion in economic loss

Alterations to the River's Course

One major effect of an earthquake is that it can change the course of a river. This happens when debris from the earthquake blocks the river's flow forcing the water to find a new path. This alteration can lead to flooding and other significant changes in the landscape.

Fountains of Mud

Earthquakes can force mud and boiling water to the surface. After the 1934 Bihar earthquake, fields were covered in knee-deep mud.

Fire Risks

Earthquakes damage gas pipelines and electric systems. This makes it much harder to put out fires caused by the earthquake.

Mitigation Measures for Earthquakes in India
The National Center for Seismology
Governmental organisations receive earthquake monitoring and hazard reports from a department of the Ministry of Earth Sciences. There are three divisions in it: Geophysical Observation System, Earthquake Hazard and Risk Assessment, and Earthquake Monitoring and services.

National Earthquake Risk Mitigation Project (NERMP)
Improving both the non-structural and structural parts of earthquake safety programs helps reduce risks in high-danger areas. In places with strong earthquakes, important safety measures are implemented. The NDMA which is in charge of the project, has made a detailed project report (DPR).

National Building Code (NBC)
It is a detailed building code and national regulation that sets rules for construction across the country. The Planning Commission first published it in 1970, and it was updated in 1983. There were three major changes: two in 1987 and one in 1997. The National Building Code of India 2005 (NBC 2005) replaced the old version. It focuses on addressing natural disaster challenges and using the best international practices.

Building Materials & Technology Promotion Council (BMTPC)
It works on projects to strengthen important buildings and raise awareness among people and government organizations. The goal is to help the public and policymakers reduce the risk to many existing public and private buildings

NDMA Guidelines for Earthquakes
In 2007, the NDMA released detailed earthquake guidelines. These rules tell State Governments, Central Ministries, and Departments what to do to make disaster management plans focused on earthquake risk. The guidelines are based on six main principles

The building of new structures that is earthquake-resistant.
Retrofitting and selective seismic strengthening of existing structures.
Enforcement and regulation.

Preparation and awareness.
Building capacity;
Emergency reaction.
Biggest Earthquakes in India
The devastating Bhuj earthquake happened on 26 January 2001 near the Pakistani border in Gujarat India. The largest earthquake in India with a magnitude of 8.6, occurred in the India-China region on 15 August 1950. It caused the death of 1530 people due to tectonic plates shifting 30 km deep.

Earthquake in the Indian Ocean
The 2004 Indian Ocean earthquake and tsunami important points

Magnitude: Between 9.1 and 9.3 on the Richter scale, making it one of the largest earthquakes ever recorded.
Duration: Faulting lasted between 8.3 and 10 minutes, unusually long for an earthquake of this magnitude.
Aftershocks: Numerous aftershocks continued for 3 to 4 months after the initial earthquake.
Energy Release: The earthquake released a massive amount of energy, causing significant geological effects.
Earth's Axis Shift: It is believed that the earthquake caused a slight shift in the Earth's axis due to the redistribution of mass.
Tsunami Generation: The seismic activity caused vertical movement of the seafloor, displacing a large volume of water and triggering a tsunami.
Impact: Indonesia was the first and hardest-hit country due to its proximity to the epicenter.
Casualties: Approximately 170000 people lost their lives, making it one of the deadliest natural disasters in recorded history

Human adjustment to and perception of earthquakes can influence how people prepare for and mitigate the effects of earthquakes:

- Risk perception

  Risk perception is influenced by factors such as fear of future earthquakes, expected damage, and expected breakdown of supplies.

- Adaptation behavior

  Adaptation behavior is how people respond to earthquake disasters. Some people may take available approaches, while others may not.

- Policy efficacy
  Policy efficacy is a person's belief in their ability to understand and engage in earthquake risk management activities.

Here are some ways to prepare for and mitigate the effects of earthquakes:

- Create an emergency plan: Make a family emergency communications plan with an out-of-state contact and a meeting place in case you get separated.
- Prepare a supply kit: Include enough food and water for several days, a flashlight, a fire extinguisher, and a whistle.

- Secure furniture: Use brackets or seismic restraints to secure tall furniture and appliances to wall framing.
- Hang heavy items away from beds: Hang heavy items like pictures and mirrors away from beds, couches, and anywhere people sleep or sit.
- Build safer structures: Build safer structures and identify hazards.
- Educate the population: Provide education on earthquake safety.

Lecture 4

What is Cyclone?

In meteorology, the term cyclone is defined as

A system of winds that are rotating inwards to an area of low barometric pressure, such that in the Northern Hemisphere it is anticlockwise and in the Southern Hemisphere it is clockwise circulation.

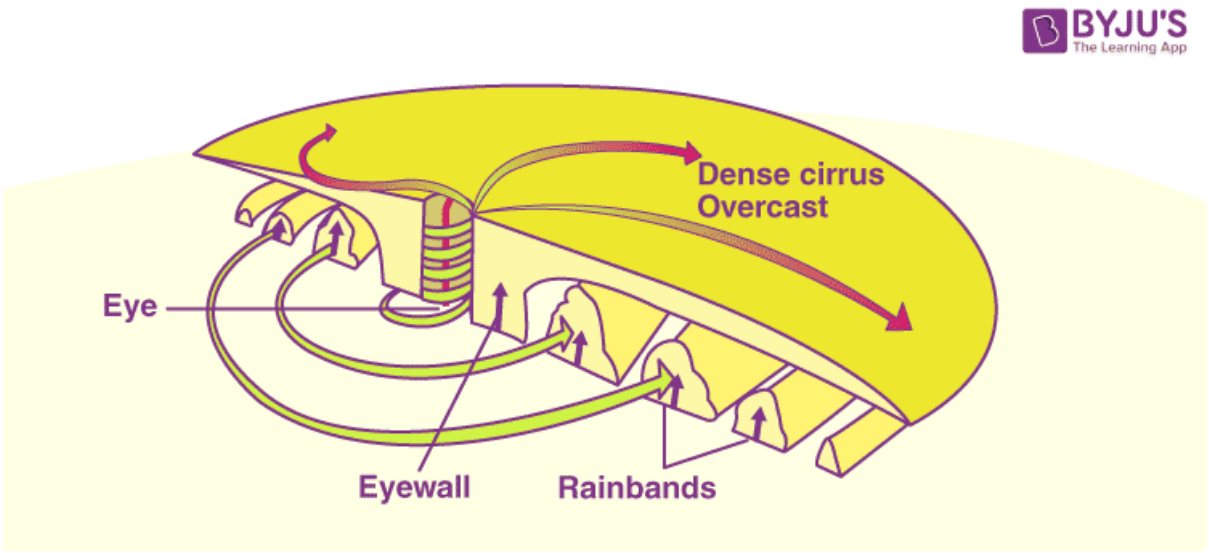
Cyclones are formed with an enormous amount of energy from the ocean to the atmosphere. According to studies, every year there are 70 to 90 cyclonic systems developed across the globe. The surface winds get spiralled towards the low-pressure system due to the Coriolis force. The cyclonic systems are not formed in the equatorial regions because the Coriolis force is negligible between the latitudes 5 degrees north and 5 degrees south.

How are Cyclones Formed?

The formation of cyclones takes place in low-pressure areas. The vulnerability of the place where the cyclone strikes depend on the topography, intensity and frequency of the cyclone.

There are 6 factors that can be held responsible for the formation of the cyclone:

- Ample amount of warm temperature at the surface of the sea.
- Instability in the atmosphere.
- How the Coriolis force is impacting the area so that low-pressure area can be created.
- When the humidity is high in the lower to middle levels of the troposphere.
- Disturbance in the pre-existing low-level area.
- When the vertical wind shear is low.



In the above diagram, we see how cyclones are formed. The rising of warm air is shown using the green lines while the cool air sinking is shown using the red arrows.

- A cyclone is formed when the warm, moist air rises upward over the ocean. As this air moves up, there is a formation of a low-pressure area below.
- Now the low-pressure area is filled with the high-pressure air from the surroundings. Again, the next batch of cool air gets warm and moist over the ocean moving upward. This once again results in the formation of a low-pressure area. The cycle continues.
- This continuous cycle results in the formation of clouds in the air. More clouds are formed as the water from the ocean evaporates.
- This leads to the formation of the storm system. As the storm system rotates faster, an eye is formed in the centre. The eye of the storm is considered to be the calm and clear part. The air pressure in the eye of the storm is low.
- Depending on the strength of the winds produced, the cyclones are divided into different classes.

What are the Types of Cyclones?

There are 4 types of cyclones and they are:

- Tropical cyclone
- Polar cyclone
- Mesocyclone
- Extratropical cyclone

Tropical Cyclone

The regions where the tropical cyclone occurs are over the tropical ocean. Hurricanes and typhoons are the two types of tropical cyclone. Atlantic and Northeast Pacific are the regions where hurricanes are found. Whereas, in the Northwest Pacific, typhoons are found. A tropical cyclone is divided into 5 categories depending on the intensity and the wind speed.

Polar Cyclone

The polar cyclone is also known as the “Arctic hurricane” in the Northern Hemisphere. This is due to their energy sources. The heat is transferred from water to air and the latent heat is released in the form of cloud condensation. The forecast of the polar cyclone is difficult as they take less than 24 hours, and they are formed quickly. They are formed over the Arctic and Antarctic seas.

Mesocyclone

The mesocyclone is considered to be one of the strong thunderstorms. The mesocyclone is found within the convective storm in the form of a vortex of air. The air rises and rotates along the vertical axis. The direction of this air and the low-pressure system are the same in the given hemisphere. The mesocyclone is accompanied by the rotating air within the thunderstorm.

Extratropical Cyclone

The cyclones that are formed in the middle latitudes along the frontal boundaries are either known as extratropical cyclones or mid-latitude cyclones. The winds of the extratropical cyclone are weaker however,

the temperature gradients are sharp. The nor'easter is an example of an extratropical cyclone that impacts the east coast of the United States.

Tropical cyclones are compact, circular storms that form over ocean basins in lower latitudes. They are characterized by strong winds, a low-pressure center, and a spiral arrangement of thunderstorms. Depending on their location and strength, tropical cyclones are also known as hurricanes, typhoons, tropical storms, and cyclones.

A tropical cyclone is a weather phenomenon that is essentially a rapidly rotating storm system with characteristics such as a low-pressure center, strong winds and thunderstorms that produce heavy rain, among others.

The topic, 'Tropical Cyclone' is important for IAS Exam as it is an important section of Geography which comes under both Prelims and UPSC Mains GS-I. The article will about in detail about the Tropical Cyclones, their characteristics and their formation.

What are the characteristics of a tropical cyclone?

Tropical cyclones or cyclones, in general, have different terms and names depending on their location and strength.

- These include 'hurricanes' in the North Atlantic Ocean and eastern North Pacific region,
- While being referred to as typhoons in the western North Pacific region.
- The nomenclature in the Southern Pacific and the Indian Ocean severe tropical cyclones or simply cyclones.

Tropical cyclones are compact, circular winds with a diameter of 320km. Its winds swirl around a central region that has low atmospheric pressure. The rotation of the winds is largely driven by the low-pressure centre and by the rotation of the Earth.

Thus cyclones rotate in a counterclockwise direction in the Northern Hemisphere and in a clockwise direction while in the Southern Hemisphere. Unlike popular belief, cyclones do not cause massive damage to underwater coral reefs. To know how these tropical cyclones are formed, read below.

The passage of a tropical cyclone over the ocean causes the upper layers of the ocean to cool substantially, which can influence subsequent cyclone development. This cooling is primarily caused by wind-driven mixing of cold water from deeper in the ocean with the warm surface waters. This effect results in a negative feedback process that can inhibit further development or lead to weakening. Additional cooling may come in the form of cold water from falling raindrops (this is because the atmosphere is cooler at higher altitudes). Cloud cover may also play a role in cooling the ocean, by shielding the ocean surface from direct sunlight before and slightly after the storm passage.

How are Tropical Cyclones formed?

In the tropics, there is a narrow zone of low pressure which stretches across the equator. The winds on the north side of this zone blow from the north-east (the north-east trades) and on the southern side blow from the south-east (south-east trades).

The low-pressure area is heated over the warm tropical ocean which leads to the rise of air from this are in discrete parcels, ultimately causing the formation of thunderstorms. This creates a flow of very warm, moist, rapidly rising air, leading to the development of a centre of low pressure, or depression, at the surface.

There are various trigger mechanisms required to transform these cloud clusters into a tropical cyclone which is at least a trigger for bad weather. These trigger mechanisms depend on several conditions being 'right' at the same time. **The most influential factors are**:

1. A source of warm, moist air derived from tropical oceans with sea surface temperatures normally in the region of, or in excess, of 27 °C;
2. Winds near the ocean surface blowing from different directions converging and causing air to rise and storm clouds to form;
3. Winds which do not vary greatly with height – known as low wind shear. This allows the storm clouds to rise vertically to high levels;
4. Sufficient distance from the equator to provide spin or twist.

La Nina, a weather pattern most commonly found in the Southern Pacific region also cause cyclones depending upon the weather condition at the time.

Where and when do tropical cyclones occur?

Many tropical cyclones eventually drift far enough from the equator to move into areas dominated by westerly winds (found in the middle latitudes). These winds tend to reverse the direction of the tropical cyclone to an eastward path. As the tropical cyclone moves polewards it picks up forward speed and may reach 30 m.p.h. or more. **An average tropical cyclone can travel about 300 to 400 miles a day, or about 3,000 miles before it dies out.**

Tropical cyclones which occur in the Atlantic region usually comprise less than 15% of global tropical cyclone activity. Tropical cyclones also occur in various parts of the Pacific Ocean and can affect coastal regions of Mexico, south-east Asia, north-east Australia and the South Pacific islands. Those that form in the Indian Ocean can affect India, Bangladesh, north-west Australia, some parts of East Africa and Indian Ocean islands such as Mauritius and Madagascar.

Tropical cyclones can cause localized damaging winds, such as downbursts, gusts, and swirls. These winds can be especially strong when a tropical cyclone makes landfall, as friction with the surface increases turbulence.

Here are some other effects of tropical cyclones:

- Tornadoes: Tropical cyclones can generate tornadoes.

- Stagnant water: Stagnant water can spread disease.
- Infrastructure damage: Damaged buildings, electrical lines, telephone towers, and water pipes can hamper rescue and clean-up efforts.
- Price spikes: Prices of food and other necessities can spike.
- Water-borne diseases: Water-borne diseases can be a long-term hazard.

Tropical cyclones can cause a variety of destruction, including:

- Damage to buildings: Well-built homes can experience major damage, while others can be destroyed completely.
- Tree damage: Trees can be snapped or uprooted, blocking roads and creating fuel for wildfires.
- Flooding: Storm surges can cause severe flooding, which can destroy buildings, erode beaches and dunes, and damage roads and bridges.
- Power outages: Power outages can last for weeks or months.
- Loss of life: Tropical cyclones can cause high death tolls.
- Soil erosion: Tropical cyclones can cause soil erosion.

Cyclones, also known as hurricanes or typhoons depending on location, are formed primarily due to warm ocean waters with high levels of evaporation, creating atmospheric instability which leads to rising air and the formation of low-pressure systems; their distribution is concentrated in specific tropical regions, with humans adjusting by building protective structures, developing early warning systems, and implementing evacuation plans, while perceptions of cyclone risks can vary based on cultural factors and past experiences, and mitigation efforts focus on land-use planning, coastal defenses, and community preparedness strategies.

Causes of Cyclones:

- Warm Ocean Waters:

  The primary factor, where heat energy from the ocean is transferred to the atmosphere through evaporation, creating the necessary moisture for cloud formation.
- Atmospheric Instability:

  A condition where warm, moist air rapidly rises, causing convection and low pressure at the surface.
- Coriolis Force:

  This force, caused by Earth's rotation, deflects the wind flow, creating a spinning motion characteristic of cyclones.
- Convergence of Winds:
  When winds from different directions meet, they converge and rise, contributing to cyclone development.

Distribution of Cyclones:

- Tropical Regions:

Cyclones primarily form in the tropics due to the presence of warm ocean waters, with major cyclone basins including the Atlantic, Pacific, Indian, and South Pacific Oceans.

- Seasonal Variation:
  Cyclone activity varies depending on the season, with peak occurrences during specific months in different regions.

Human Adjustments to Cyclones:

- Coastal Infrastructure:

  Building seawalls, embankments, and elevated structures to protect against storm surges.

- Evacuation Plans:

  Establishing designated evacuation routes and shelters to move people away from coastal areas before a cyclone hits.

- Early Warning Systems:

  Utilizing weather monitoring and forecasting technology to issue timely cyclone warnings to communities.

- Community Preparedness:
  Educating communities on cyclone risks, first aid, and emergency response procedures.

Perception of Cyclone Risks:

- Cultural Factors:

  Past experiences with cyclones, local beliefs, and community norms can influence how people perceive cyclone risks.

- Socioeconomic Status:

  Communities with limited resources may be more vulnerable to cyclone impacts due to lack of access to information and preparedness measures.

- Trust in Authorities:
  Confidence in the accuracy of weather forecasts and government response can impact preparedness behaviors.

Mitigation Strategies for Cyclones:

- Land Use Planning:

  Avoiding construction in high-risk areas like floodplains and coastal dunes.

- Mangrove Restoration:

  Planting mangroves along coastlines to act as natural barriers against storm surges.

- Building Codes:

  Implementing stricter building codes to ensure structures can withstand cyclone winds

- Disaster Risk Reduction (DRR):
  Integrating cyclone risk reduction into development plans, including early warning systems, community awareness campaigns, and infrastructure improvements.

- Tertiary hazards: Diseases caused by stagnant water, water poisoning, and increased prices of goods and resources.

Tropical cyclones can also affect food security, health, and biodiversity. For example, tropical cyclones can destroy coral reefs and defoliate tree canopies, which can affect the food supply.

Lecture 5

A **cold wave** (known in some regions as a **cold snap**, **cold spell** or **Arctic Snap**) is a weather phenomenon that is distinguished by a cooling of the air. Specifically, as used by the U.S. National Weather Service, a cold wave is a rapid fall in temperature within a 24-hour period requiring substantially increased protection to agriculture, industry, commerce, and social activities. The precise criteria for a cold wave are the rate at which the temperature falls, and the minimum to which it falls. This minimum temperature is dependent on the geographical region and time of year.[1]

In the United States, a *cold spell* is defined as the national average high temperature dropping below 20 °F (−7 °C).[2] A cold wave of sufficient magnitude and duration may be classified as a **cold air outbreak** (**CAO**).

Effects

A cold wave can cause death and injury to livestock and wildlife. Exposure to cold mandates greater caloric intake for all animals, including humans, and if a cold wave is accompanied by heavy and persistent snow, grazing animals may be unable to reach needed food and die of hypothermia or starvation. They often necessitate the purchase of foodstuffs to feed livestock at considerable cost to farmers.

Cold spells are associated with increased mortality rates in populations around the world.[5] Both cold waves and heat waves cause deaths, though different groups of people may be susceptible to different weather events.[6] More temperature-attributable deaths occur during a cold wave than in a heat wave,[7][8] though the mortality rate is higher in undeveloped regions of the world. Extreme winter cold often causes poorly insulated water pipelines and mains to freeze. Even some poorly protected indoor plumbing ruptures as water expands within them, causing much damage to property and costly insurance claims. Demand for electrical power and fuels rises dramatically during such times, even though the generation of electrical power may fail due to the freezing of water necessary for the generation of hydroelectricity. Some metals may become brittle at low temperatures. Motor vehicles may fail when antifreeze fails or motor oil gels, producing a failure of the transportation system.

Fires become even more of a hazard during extreme cold. Water mains may break and water supplies may become unreliable, making firefighting more difficult. The air during a cold wave is typically denser and thus contains more oxygen, so when air that a fire draws in becomes unusually cold it is likely to cause a more intense fire.[*citation needed*] However, snow may stop spreading of fires, especially wildfires.

Winter cold waves that are not considered cold in some areas, but cause temperatures significantly below average for an area, are also destructive. Areas with subtropical climates may recognize a cold wave at higher temperatures than other, colder areas of the globe. The cold wave may be recognized at barely freezing temperatures, as these are still unusually cold for the region, and plant and animal life will be less tolerant of such cold. The same winter temperatures that one associates with the norm for Colorado, Ohio, or Bavaria are catastrophic to crops in places like Florida, California, or parts of South America that grow fruit and vegetables in winter.

Cold waves that bring unexpected freezes and frosts during the growing season in mid-latitude zones can kill plants during the early and most vulnerable stages of growth, resulting in crop failure as plants are killed before they can be harvested economically. Such cold waves have caused famines. At times as deadly to plants as drought, cold waves can leave land in danger of later brush and forest fires that consume dead biomass. One extreme was the so-called Year Without a Summer of 1816, one of several years during the 1810s in which numerous crops failed during freakish summer cold snaps after volcanic eruptions that reduced incoming sunlight.

Recent research suggests a possible link between cold waves in North America and extratropical cyclogenesis over the East Atlantic.[9] These may be connected by large-scale atmospheric circulation patterns. [10] Examples include Rossby wave propagation from the North Pacific or an upper-

level anticyclone west of Greenland.[11] In Europe, the advection of cold air masses from the northeast emerges as a potential precursor signal for the majority of cold waves, significantly affecting the energy, health, and agricultural sectors of the continent.[12]

Countermeasures

In some places, such as Siberia, extreme cold requires that fuel-powered machinery intended to be used occasionally must be run continually. Internal plumbing can be wrapped, and persons can often run water continuously through pipes. Energy conservation, difficult as it is in a cold wave, may require such measures as collecting people (especially the poor and elderly) in communal shelters. Even the homeless may be arrested and taken to shelters, only to be released when the hazard abates.[13] Hospitals can prepare for the admission of victims of frostbite and hypothermia; schools and other public buildings can be converted into shelters.

People can stock up on food, water, and other necessities before a cold wave. Some may even choose to migrate to places of milder climates, at least during the winter. Suitable stocks of forage can be secured before cold waves for livestock, and livestock in vulnerable areas might be shipped from affected areas or even slaughtered. Smudge pots can bring smoke that prevents hard freezes on a farm or grove. Vulnerable crops may be sprayed with water that will paradoxically protect the plants by freezing and absorbing the cold from surrounding air.

Most people can dress appropriately and can layer their clothing should they need to go outside or should their heating fail. They can also stock candles, matches, flashlights, and portable fuel for cooking and wood for fireplaces or wood stoves, as necessary. However, caution should be taken as the use of charcoal fires for cooking or heating within an enclosed dwelling is extremely dangerous due to carbon monoxide poisoning. Adults must remain aware of the exposure that children and the elderly have to cold.

A **heat wave**[1] or **heatwave**,[2] sometimes described as **extreme heat**, is a period of abnormally hot weather.[3]:2911 Definitions vary but are similar.[4] A *heat wave* is usually measured relative to the usual climate in the area and to normal temperatures for the season.[3]:2911 Temperatures that humans from a hotter climate consider normal, can be regarded as a heat wave in a cooler area. This would be the case if the warm temperatures are outside the normal climate pattern for that area.[5] High humidity often occurs during heat waves as well. This is especially the case in oceanic climate countries. Heat waves have become more frequent, and more intense over land, across almost every area on Earth since the 1950s, the increase in frequency and duration being caused by climate change.[6][7]

Heat waves form when a high-pressure area in the upper atmosphere strengthens and remains over a region for several days up to several weeks.[8] This traps heat near the earth's surface. It is usually possible to forecast heat waves, thus allowing the authorities to issue a warning in advance.

Heat waves have an impact on the economy. They can reduce labour productivity, disrupt agricultural and industrial processes and damage infrastructure.[9][10] Severe heat waves have caused catastrophic crop failures and thousands of deaths from hyperthermia. They have increased the risk of wildfires in areas with drought. They can lead to widespread electricity outages because more air conditioning is used. A heat wave counts as extreme weather. It poses danger to human health, because heat and sunlight overwhelm the thermoregulation in humans.

There are several definitions of heat waves:

- The IPCC defines *heatwave* as "a period of abnormally hot weather, often defined with reference to a relative temperature threshold, lasting from two days to months."[11][3]:2911
- A definition based on the *Heat Wave Duration Index* is that a heat wave occurs when the daily maximum temperature of more than five consecutive days exceeds the average maximum temperature by 5 °C (9 °F), the normal period being 1961–1990.[12] The same definition is used by the World Meteorological Organization.[13]

- A definition from the *Glossary of Meteorology* is:[14] "A period of abnormally and uncomfortably hot and usually humid weather."

Definitions by country

Denmark defines a national heat wave (*hedebølge*) as a period of at least 3 consecutive days in which the average maximum temperature across more than half the country exceeds 28 °C (82.4 °F). The Danish Meteorological Institute also has a definition for a "warmth wave" (*varmebølge*). It defines this using the same criteria for a 25 °C (77.0 °F) temperature.[15] Sweden defines a heat wave as at least five days in a row with a daily high exceeding 25 °C (77.0 °F).[16]

In Greece, the Hellenic National Meteorological Service defines a heat wave as occurring over three consecutive days with temperatures at 39 °C (102 °F) or higher. In the same period the minimum temperature is 26 °C (79 °F) or more. During this period, there are either no winds or only weak winds. These conditions occur in a broad area.

The Netherlands defines a heat wave as a period of at least five consecutive days in which the maximum temperature in De Bilt exceeds 25 °C (77 °F). During this period the maximum temperature in De Bilt must exceed 30 °C (86 °F) for at least three days. Belgium also uses this definition of a heat wave with Ukkel as a reference point. So does Luxembourg.

In the United Kingdom, the Met Office operates a Heat Health Watch system. This places each Local Authority region into one of four levels. Heat wave conditions occur when the maximum daytime temperature and minimum nighttime temperature rise above the threshold for a particular region. The length of time above that threshold determines the level. Level 1 represents normal summer conditions. Level 2 occurs when there is a 60% or higher risk that the temperature will be above the threshold levels for two days and the intervening night. Level 3 arises when the temperature has been above the threshold for the preceding day and night, and there is a 90% or higher chance that it will stay above the threshold in the following day. Level 4 is triggered if conditions are more severe than those of the preceding three levels. Each of the first three levels gives rise to a particular state of readiness and response by the social and health services. Level 4 involves a more widespread response.[17] The threshold for a heat wave occurs when there are at least three days above 25 °C (77 °F) across much of the country. Greater London has a threshold of 28 °C (82 °F).[18]

Other regions

In the United States, definitions also vary by region. They usually involve a period of at least two or more days of excessively hot weather.[19] In the Northeast, a heat wave is typically when the temperature reaches or exceeds 90 °F (32.2 °C) for three consecutive days. This is not always the case. This is because the high temperature ties in with humidity levels to determine a heat index threshold.[20] The same does not apply to drier climates. A heat storm is a Californian term for an extended heat wave. Heat storms occur when the temperature reaches 100 °F (37.8 °C) for three or more consecutive days over a wide area (tens of thousands of square miles).[21] The National Weather Service issues heat advisories and excessive heat warnings when it expects unusual periods of hot weather.

In Adelaide, South Australia, a heat wave is five consecutive days at or above 35 °C (95 °F), or three consecutive days at or over 40 °C (104 °F).[22] The Australian Bureau of Meteorology defines a heat wave as three or more days of unusual maximum and minimum temperatures.[23] Before this new Pilot Heatwave Forecast there was no national definition for heat waves or measures of heat wave severity.[23]

It is possible to compare heat waves in different regions of the world with different climates thanks to a general indicator that appeared in 2015.[27] With these indicators, experts estimated heat waves at the global scale from 1901 to 2010. They found a substantial and sharp increase in the number of affected areas in the last two decades.[28]

One study in 2021 investigated 13,115 cities. It found that extreme heat exposure of a wet bulb globe temperature above 30 Celsius tripled between 1983 and 2016, and if the effect of population growth

(increasing the urban heat island effect) during those years is excluded, the exposure increased a further 50%. The researchers compiled a comprehensive list of past urban extreme heat events.[29][30]

Causes

Heat waves form when a high pressure area at an altitude of 10,000–25,000 feet (3,000–7,600 metres) strengthens and remains over a region for several days and up to several weeks.[8] This is common in summer in both the Northern and Southern Hemispheres. This is because the jet stream 'follows the sun'. The high pressure area is on the equator side of the jet stream in the upper layers of the atmosphere.

Weather patterns are generally slower to change in summer than in winter. So, this upper level high pressure also moves slowly. Under high pressure, the air sinks toward the surface. It warms and dries adiabatically. This inhibits convection and prevents the formation of clouds. A reduction of clouds increases the shortwave radiation reaching the surface. A low pressure area at the surface leads to surface wind from lower latitudes that brings warm air, enhancing the warming. The surface winds could also blow from the hot continental interior towards the coastal zone. This would lead to heat waves on the coast. They could also blow from high towards low elevations. This enhances the subsidence or sinking of the air and therefore the adiabatic warming.[31][32]

In the eastern regions of the United States a heat wave can occur when a high pressure system originating in the Gulf of Mexico becomes stationary just off the Atlantic Seaboard. Hot humid air masses form over the Gulf of Mexico and the Caribbean Sea. At the same time hot dry air masses form over the desert Southwest and northern Mexico. The southwest winds on the back side of the high continue to pump hot, humid Gulf air northeastwards. This results in a spell of hot and humid weather for much of the eastern United States and into southeastern Canada.[33]

In the Western Cape Province of South Africa, a heat wave can occur when the low-pressure area offshore and the high-pressure area inland combine to form a bergwind. The air warms as it descends from the Karoo interior. The temperature will rise about 10 Celsius from the interior to the coast. Humidity is usually very low. The temperature can be over 40 Celsius in summer. The highest temperature recorded in South Africa (51.5 Celsius) occurred one summer during a berg wind along the Eastern Cape coastline.[34][35]

The level of soil moisture can intensify heat waves in Europe.[36][37] Low soil moisture leads to a number of complex feedback mechanisms. These in turn can result in increased surface temperatures. One of the main mechanisms is reduced evaporative cooling of the atmosphere.[36] When water evaporates, it consumes energy. So, it will lower the surrounding temperature. If the soil is very dry, then incoming radiation from the sun will warm the air. But there will be little or no cooling effect from moisture evaporating from the soil.

Climate change

Heatwaves over land have become more frequent and more intense in almost all world regions since the 1950s, due to climate change. Heat waves are more likely to occur simultaneously with droughts. Marine heatwaves are twice as likely as they were in 1980.[38] Climate change will lead to more very hot days and fewer very cold days.[39]:7 There are fewer cold waves.[40]:8

Experts can often attribute the intensity of individual heat waves to global warming. Some extreme events would have been nearly impossible without human influence on the climate system. A heatwave that would occur once every ten years before global warming started now occurs 2.8 times as often. Under further warming, heatwaves are set to become more frequent. An event that would occur every ten years would occur every other year if global warming reaches 2 °C (3.6 °F).[41]

Impacts on human health

Heat-related health effects for vulnerable humans

Heat illness is a spectrum of disorders due to increased body temperature. It can be caused by either environmental conditions or by exertion. It includes minor conditions such as heat cramps, heat syncope, and heat exhaustion as well as the more severe condition known as heat stroke.[42] It can affect any or all anatomical systems.[43] Heat illnesses include:[44][45] heat stroke, heat exhaustion, heat syncope, heat edema, heat cramps, heat rash, heat tetany.

Prevention includes avoiding medications that can increase the risk of heat illness, gradual adjustment to heat, and sufficient fluids and electrolytes.[46][47]

Vulnerable people with regard to heat illnesses include people with low incomes, minority groups, women (in particular pregnant women), children, older adults (over 65 years old), people with chronic diseases, disabilities and co-morbidities.[48]:13 Other people at risk include those in urban environments (due to the urban heat island effect), outdoor workers and people who take certain prescription drugs.[48] Exposure to extreme heat poses an acute health hazard for many of the people deemed as vulnerable.[48][49]

Climate change increases the frequency and severity of heatwaves and thus heat stress for people. Human responses to heat stress can include heat stroke and hyperthermia. Extreme heat is also linked to low quality sleep, acute kidney injury[50][51] and complications with pregnancy. Furthermore, it may cause the deterioration of pre-existing cardiovascular and respiratory disease.[52]:1624 Adverse pregnancy outcomes due to high ambient temperatures include for example low birth weight and pre-term birth.[52]:1051 Heat waves have also resulted in epidemics of chronic kidney disease (CKD).[53][54] Prolonged heat exposure, physical exertion, and dehydration are sufficient factors for the development of CKD.[53][54]

Mortality

Category	Risk of Heat-Related Impacts
Green 0	Little to no risk from expected heat.
Yellow 1	Minor - This level of heat affects primarily those individuals extremely sensitive to heat, especially when outdoors without effective cooling and/or adequate hydration.
Orange 2	Moderate - This level of heat affects most individuals sensitive to heat, especially those without effective cooling and/or adequate hydration. Impacts possible in some health systems and in heat-sensitive industries.
Red 3	Major - This level of heat affects anyone without effective cooling and/or adequate hydration. Impacts likely in some health systems, heat-sensitive industries and infrastructure.
Magenta 4	Extreme - This level of rare and/or long-duration extreme heat with little to no overnight relief affects anyone without effective cooling and/or adequate hydration. Impacts likely in most health systems, heat-sensitive industries and infrastructure.

The National Weather Service risk categories for *NWS HeatRisk*

This section is an excerpt from Effects of climate change on human health § Heat-related mortality.[edit]

Health experts warn that "exposure to extreme heat increases the risk of death from cardiovascular, cerebrovascular, and respiratory conditions and all-cause mortality. Heat-related deaths in people older than 65 years reached a record high of an estimated 345 000 deaths in 2019".[48]:9 More than 70,000 Europeans died as a result of the 2003 European heat wave.[55] Also more than 2,000 people died in Karachi, Pakistan in June 2015 due to a severe heat wave with temperatures as high as 49 °C (120 °F).[56][57]

Due to climate change temperatures rose in Europe and heat mortality increased. From 2003–12 to 2013–22 alone, it increased by 17 deaths per 100,000 people, while women are more vulnerable than men.[58]

Underreporting of fatalities

The number of heat fatalities is probably highly underreported. This is due to a lack of reports and to misreporting.[59] When considering heat-related illnesses as well, actual death tolls from extreme heat may be six times higher than official figures. This is based on studies of California[60] and Japan.[61]

Part of the mortality during a heat wave may be due to short-term forward mortality displacement. In some heat waves there is a decrease in overall mortality in the weeks after a heat wave. These compensatory reductions in mortality suggest that heat affects people who would have died anyway, and brings their deaths forward.[62]

Social institutions and structures influence the effects of risks. This factor can also help explain the underreporting of heat waves as a health risk. The deadly French heat wave in 2003 showed that heat wave dangers result from a combination of natural and social factors.[63] Social invisibility is one such factor. Heat-related deaths can occur indoors, for instance among elderly people living alone. In these cases it can be challenging to assign heat as a contributing factor.[64]

The heat index in the table above is a measure of how hot it feels when relative humidity is factored with the actual air temperature.

Psychological and sociological effects

Excessive heat causes psychological stress as well as physical stress. This can affect performance. It may also lead to an increase in violent crime.[65] High temperatures are associated with increased conflict between individuals and at the social level. In every society, crime rates go up when temperatures go up. This is particularly the case with violent crimes such as assault, murder and rape. In politically unstable countries, high temperatures can exacerbate factors that lead to civil war.[66]

High temperatures also have a significant effect on income. A study of countries in the United States found that the economic productivity of individual days declines by about 1.7 percent for each degree Celsius above 15 °C (59 °F).[67]

Surface ozone (air pollution)

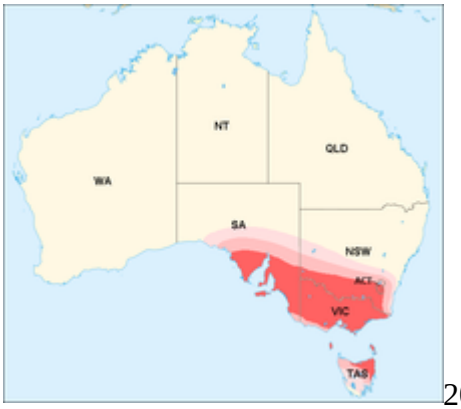
High temperatures also make the effects of ozone pollution in urban areas worse. This raises heat-related mortality during heat waves.[68] During heat waves in urban areas, ground level ozone pollution can be 20 percent higher than usual.[69]

One study looked at fine particle concentrations and ozone concentrations from 1860 to 2000. It found that the global population-weighted fine particle concentrations increased by 5 percent due to climate change. Near-surface ozone concentrations rose by 2 percent.[70]

An investigation to assess the joint mortality effects of ozone and heat during the European heat waves in 2003 concluded that these appear to be reinforce each other and increase mortality when combined.[71]

Impacts on societies

Reduced economic outputs



2009 southeastern Australia heat wave, thermal map approximate affected area shown in red

Calculations from 2022 suggest that heat waves will shrink the global economy by about 1 percent decrease by the middle of the 21st century.[72][73][74]

Heat waves often have complex effects on economies. They reduce labour productivity, disrupt agricultural and industrial processes and damage infrastructure that is not suitable for extreme heat.[9][10] In 2016, a marine heatwave in Chile and its subsequent harmful algal bloom caused $800 million (USD) in export losses for the aquaculture industry as salmon and shellfish died off.[75]

Reduced agricultural outputs

Heat waves are a big threat to agricultural production. In 2019 heat waves in the Mulanje region of Malawi involved temperatures as high as 40 °C (104 °F). This and a late rain season scorched tea leaves and reduced yields.[76]

Farmed animals

This section is an excerpt from Effects of climate change on livestock § Health impacts of heat stress.[edit]



Impacts of heat stress on livestock animals.[77]

Once the body temperature of livestock animals is 3–4 °C (5.4–7.2 °F) above normal, this soon leads to "heat stroke, heat exhaustion, heat syncope, heat cramps, and ultimately organ dysfunction". Livestock mortality rates are already known to be higher during the hottest months of the year, as well as during heatwaves. During the 2003 European heat wave, for instance, thousands of pigs, poultry, and rabbits died in the French regions of Brittany and Pays-de-la-Loire alone.[77]

Livestock can also suffer multiple sublethal impacts from heat stress, such as reduced milk production. Once the temperatures exceed 30 °C (86 °F), cattle, sheep, goats, pigs and chickens all begin to consume 3–5% less feed for each subsequent degree of temperature increase.[78] At the same time, they increase respiratory and sweating rates, and the combination of these responses can lead to metabolic

disorders. One examples is ketosis, or the rapid accumulation of ketone bodies, caused by the animal's body rapidly catabolizing its fat stores to sustain itself.[77] Heat stress also causes an increase in antioxidant enzyme activities, which can result in an imbalance of oxidant and antioxidant molecules, otherwise known as oxidative stress. Feed supplementation with antioxidants like chromium can help address oxidative stress and prevent it from leading to other pathological conditions, but only in a limited way.[79]

The immune system is also known to be impaired in heat-stressed animals, rendering them more susceptible to various infections.[77] Similarly, vaccination of livestock is less effective when they suffer from heat stress.[80] So far, heat stress had been estimated by researchers using inconsistent definitions, and current livestock models have limited correlation with experimental data.[81] Notably, since livestock like cows spend much of their day laying down, comprehensive heat stress estimation needs to take account of ground temperature as well,[82] but the first model to do so was only published in 2021, and it still tends to systematically overestimate body temperature while underestimating breathing rate.[83]

Infrastructural damage

Heat waves cause roads and highways to buckle and melt,[84] water lines to burst, and power transformers to detonate, causing fires. A heat wave can also damage railways, by buckling and kinking rails. This can slow down or delay traffic. It can even lead to cancellations of service when rails are too dangerous to traverse by trains.

Power outages

Heat waves often lead to spikes in electricity demand because there is more use of air conditioning. This can create power outages, making the problem worse. During the 2006 North American heat wave, thousands of homes and businesses went without power, especially in California. In Los Angeles, electrical transformers failed, leaving thousands without power for as long as five days.[85] The early 2009 southeastern Australia heat wave caused major power disruptions in the city of Melbourne. They left over half a million people without power as the heat wave blew transformers and overloaded a power grid.

Impacts on the environment

Wildfires

A heat wave occurring during a drought can contribute to bushfires and wildfires. This is because a drought dries out vegetation, so it is more likely to catch fire. During the disastrous heat wave that struck Europe in 2003, fires raged through Portugal. They destroyed over 3,010 square kilometres (1,160 sq mi) of forest and 440 square kilometres (170 sq mi) of agricultural land. They caused about €1 billion worth of damage.[86] High end farmlands have irrigation systems to back up crops.

Floods

Heat waves can also contribute to flooding. Because hot air is able to carry more moisture, heatwaves may be followed by extreme rainfall especially in mid-latitude regions.[87] For example, the record-breaking heat wave that afflicted Pakistan beginning in May 2022 led to glacier melt and moisture flow. These were factors in the devastating floods that began in June and claimed over 1,100 lives.[88]

Wild animals on land

Researchers have predicted that roughly 10-40% of all land vertebrate species will be affected by heat waves by 2099, depending on the amount of future greenhouse gas emissions.[89] Heatwaves present an additional form of stress and evolutionary pressure for species that already deal with habitat loss and climate change.

Species have a thermal range of tolerance that describes the temperatures where they perform best. Temperature conditions that are outside of this range may experience decreased fitness and the inability to reproduce.[90][91] The species with sufficient genetic variation will be able to ensure some individuals can survive frequent days of high temperatures in the future.[92]

Oceans

Marine heatwaves may cause mass mortality in fish populations, especially for species that are better adapted to cooler temperatures.[93] Species that have adapted to warmer temperatures may expand their range during a heatwave. These invasive species may outcompete the native species that experience higher mortality during a heatwave, which disrupts ecosystem functioning.[93] Marine heatwaves have also been correlated with negative impacts on foundation species such as coral and kelp.[94]

Options for reducing impacts on humans

A possible public health measure during heat waves is to set up air-conditioned public cooling centres. Adding air conditioning in schools provides a cooler work place.[95] But it can result in additional greenhouse gas emissions unless solar energy is used.

Policymakers, funders and researchers have created the Extreme Heat Resilience Alliance coalition under the Atlantic Council. This advocates for naming heat waves, measuring them, and ranking them to build better awareness of their impacts.[96][97]

What is lightning?

Lightning is a giant spark of electricity in the atmosphere between clouds, the air, or the ground. In the early stages of development, air acts as an insulator between the positive and negative charges in the cloud and between the cloud and the ground. When the opposite charges build up enough, this insulating capacity of the air breaks down and there is a rapid discharge of electricity that we know as lightning. The flash of lightning temporarily equalizes the charged regions in the atmosphere until the opposite charges build up again.

Lightning can occur between opposite charges within the thunderstorm cloud (intra-cloud lightning) or between opposite charges in the cloud and on the ground (cloud-to-ground lightning).

Lightning is one of the oldest observed natural phenomena on earth. It can be seen in volcanic eruptions, extremely intense forest fires, surface nuclear detonations, heavy snowstorms, in large hurricanes, and obviously, thunderstorms. .

What causes thunder?

Lightning causes thunder! Energy from a lightning channel heats the air briefly to around 50,000 degrees Fahrenheit, much hotter than the surface of the sun. This causes the air to explode outward. The huge pressure in the initial outward shock wave decreases rapidly with increasing distance and within ten yards or so has become small enough to be perceived as the sound we call thunder.

Thunder can be heard up to 25 miles away from the lightning discharge, but the frequency of the sound changes with distance from the lightning channels that produce it, because higher frequencies are more quickly absorbed by the air. Very close to lightning, the first thunder you hear is from the closest channels,which produce a tearing sound, because that thunder contains high frequencies. A few seconds later, you hear a sharp click or loud crack from lightning channels a little farther away, and several tens of seconds later the thunder from the most distant part of a flash has quieted to low frequency rumbling.

Because light travels through the air roughly a million times faster than sound does, you can use thunder to estimate the distance to lightning. Just count the number of seconds from the time you see a flash until you hear thunder. Sound travels approximately one fifth of a mile per second or one third of a kilometer per second, so dividing the number of seconds by 5 gives the number of miles to the flash and dividing by 3 gives the number of kilometers.

Where does lightning strike?

Most, if not all, lightning flashes produced by storms start inside the cloud. If a lightning flash is going to strike ground, a channel develops downward toward the surface. When it gets less than roughly a hundred yards of the ground, objects like trees and bushes and buildings start sending up sparks to meet it. When one of the sparks connects the downward developing channel, a huge electric current surges rapidly down the channel to the object that produced the spark. Tall objects such as trees and skyscrapers are more likely than the surrounding ground to produce one of the connecting sparks and so are more likely to be struck by lightning. Mountains also make good targets. However, this does not always mean tall objects will be struck. Lightning can strike the ground in an open field even if the tree line is close by.

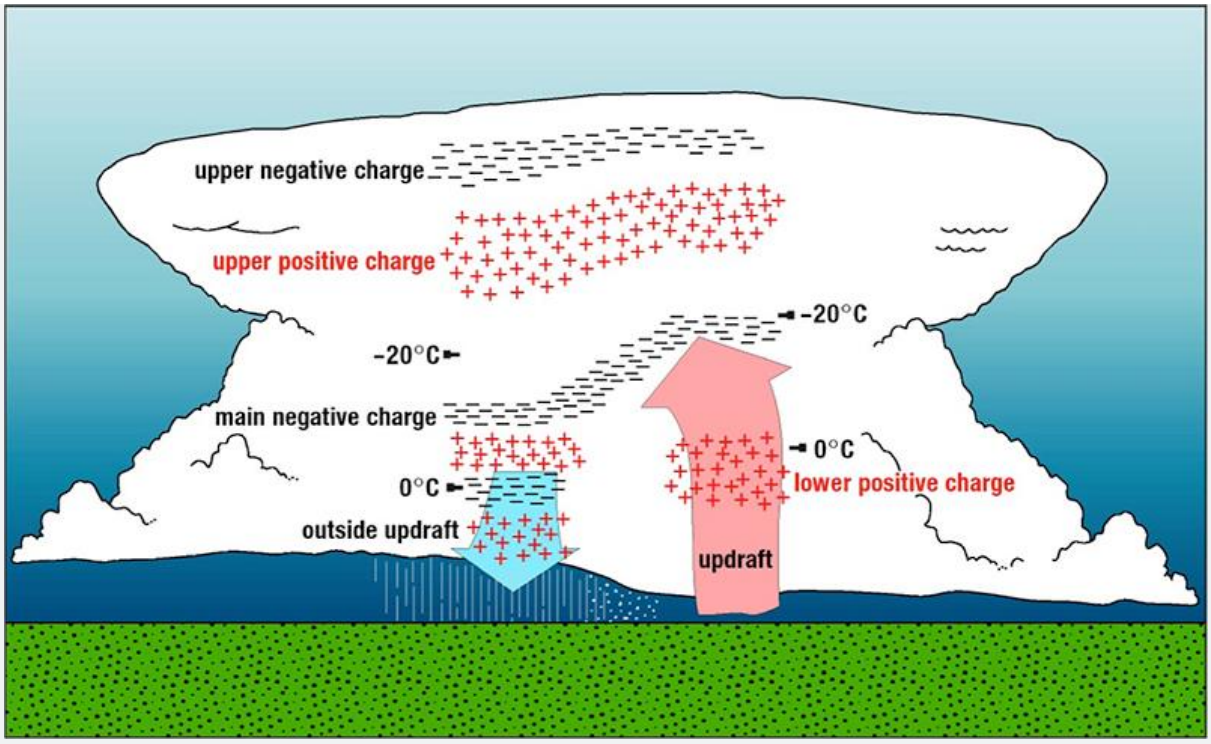
What causes lightning?

The creation of lightning is a complicated process. We generally know what conditions are needed to produce lightning, but there is still debate about exactly how a cloud builds up electrical charges, and how lightning forms. Scientists think that the initial process for creating charge regions in thunderstorms involves small hail particles called graupel that are roughly one quarter millimeter to a few millimeters in diameter and are growing by collecting even smaller supercooled liquid droplets. When these graupel particles collide and bounce off of smaller ice particles, the graupel gains one sign of charge and the smaller ice particle gains the other sign of charge. Because the smaller ice particles rise faster in updrafts than the graupel particles, the charge on ice particles separates from the charge on graupel particles, and the charge on ice particles collects above the charge on graupel.

Laboratory studies suggest that graupel gains positive charge at temperatures a little colder than 32 degrees Fahrenheit, but gains negative charge at colder temperatures a little higher in the storm. Scientists think the two largest charge regions in most storms are caused mainly by graupel carrying negative charge in the middle of the storm and ice particles carrying gained positive charge in the upper part of the storm. However, a small positive charge region often is below the main negative charge region from graupel gaining positive charge at lower, warmer altitudes. Small ice particles that have collided with negative graupel in the lower region can contribute positive charge to the middle of the storm.

A conceptual model shows the electrical charge distribution inside deep convection (thunderstorms), developed by NSSL and university scientists. In the main updraft (in and above the red arrow), there are four main charge regions. In the convective region but outside the outdraft (in and above the blue arrow), there are more than four charge regions.

How is electrical charge distributed through a thunderstorm?



Charge distribution in storm clouds [+]

A conceptual model shows the electrical charge distribution inside deep convection (thunderstorms), developed by NSSL and university scientists. In the main updraft (in and above the red arrow), there are four main charge regions. In the convective region but outside the outdraft (in and above the blue arrow), there are more than four charge regions.

What is hail?

Hail is a form of precipitation consisting of solid ice that forms inside thunderstorm updrafts. Hail can damage aircraft, homes and cars, and can be deadly to livestock and people.

What we do: *Read more about NSSL's hail research here.*

How does hail form?

Hailstones are formed when raindrops are carried upward by thunderstorm updrafts into extremely cold areas of the atmosphere and freeze. Hailstones then grow by colliding with liquid water drops that freeze onto the hailstone's surface. If the water freezes instantaneously when colliding with the hailstone, cloudy ice will form as air bubbles will be trapped in the newly formed ice. However, if the water freezes slowly, the air bubbles can escape and the new ice will be clear. The hail falls when the thunderstorm's updraft can no longer support the weight of the hailstone, which can occur if the stone becomes large enough or the updraft weakens.

Hailstones can have layers of clear and cloudy ice if the hailstone encounters different temperature and liquid water content conditions in the thunderstorm. The conditions experienced by the hailstone can change as it passes horizontally across or near an updraft. The layers, however, do not occur simply due to the hailstone going through up and down cycles inside a thunderstorm. The winds inside a thunderstorm aren't simply up and down; horizontal winds exist from either a rotating updraft, like in supercell thunderstorms, or from the surrounding environment's horizontal winds. Hailstones also do not grow from being lofted to the top of the thunderstorm. At very high altitudes, the air is cold enough (below -40°F) that all liquid water will have frozen into ice, and hailstones need liquid water to grow to an appreciable size.

How does hail fall to the ground?

Hail falls when it becomes heavy enough to overcome the strength of the thunderstorm updraft and is pulled toward the earth by gravity. Smaller hailstones can be blown away from the updraft by

horizontal winds, so larger hail typically falls closer to the updraft than smaller hail. If the winds near the surface are strong enough, hail can fall at an angle or even nearly sideways! Wind-driven hail can tear up siding on houses, break windows and blow into houses, break side windows on cars, and cause severe injury and/or death to people and animals.

How fast does hail fall?

This is a very complicated answer. The fall speed of hail primarily depends on the size of the hailstone, the friction between the hailstone and surrounding air, the local wind conditions (both horizontal and vertical), and the degree of melting of the hailstone. Early research assumed that hailstones fell like solid ice spheres and showed very high fall speeds, even for very small hailstones. However, recent research outside of NSSL using 3-D printed casts of real hailstones suspended in a vertical wind tunnel has repeatedly shown that natural hailstones fall more slowly than solid ice spheres. For small hailstones (<1-inch in diameter), the expected fall speed is between 9 and 25 mph. For hailstones that one would typically see in a severe thunderstorm (1-inch to 1.75-inch in diameter), the expected fall speed is between 25 and 40 mph. In the strongest supercells that produce some of the largest hail one might expect to see (2-inches to 4-inches in diameter), the expected fall speed is between 44 and 72 mph. However, there is much uncertainty in these estimates due to variability in the hailstone's shape, degree of melting, fall orientation, and the environmental conditions. However, it is possible for very large hailstones (diameters exceeding 4-inches) to fall at over 100 mph.

What areas have the most hail?

Although Florida has the most thunderstorms, Nebraska, Colorado, and Wyoming usually have the most hailstorms. The area where these three states meet – "hail alley" – averages seven to nine hail days per year. Other parts of the world that have damaging hailstorms include China, Russia, India and northern Italy.

When viewed from the air, it is evident that hail falls in paths known as hail swaths. These occur as storms move while the hail is falling out They can range in size from a few acres to an area 10 miles wide and 100 miles long. Some storms, instead of producing large hail, instead produce copious amounts of small hail. Storms like these have produced hail drifts that, when captured in clogged drainage channels, formed piles of hail several feet deep. Hail that completely covers roadways is especially hazardous because if deep enough, a vehicle's tires may not touch the roadway at all, with the vehicle instead driving on the hail, which acts exactly like an icy roadway in the winter.

How large can hail get?

The largest hailstone recovered in the United States fell in Vivian, South Dakota, on June 23, 2010, with a diameter of 8 inches and a circumference of 18.62 inches. It weighed 1 lb 15 oz.

Estimating Hail Size

Hail size is often estimated by comparing it to a known object. Most hailstorms are made up of a mix of different sizes, and only the very largest hail stones pose serious risk to people caught in the open. When reporting hail, estimates comparing the hail to a known object with definite size are good, but measurements using a ruler, calipers, or a tape measure are best.

- Pea = 1/4 inch diameter
- Mothball = 1/2 inch diameter
- Penny = 3/4 inch diameter
- Nickel = 7/8 inch
- Quarter = 1 inch — hail quarter size or larger is considered severe
- Ping-Pong Ball = 1 1/2 inch
- Golf Ball = 1 3/4 inches
- Tennis Ball = 2 1/2 inches
- Baseball = 2 3/4 inches
- Tea cup = 3 inches
- Softball = 4 inches
- Grapefruit = 4 1/2 inches

Lecture No 6

A **flood** is an overflow of water (or rarely other fluids) that submerges land that is usually dry.[1] In the sense of "flowing water", the word may also be applied to the inflow of the tide. Floods are of significant concern in agriculture, civil engineering and public health. Human changes to the environment often increase the intensity and frequency of flooding. Examples for human changes are land use changes such as deforestation and removal of wetlands, changes in waterway course or flood controls such as with levees. Global environmental issues also influence causes of floods, namely climate change which causes an intensification of the water cycle and sea level rise.[2]:1517 For example, climate change makes extreme weather events more frequent and stronger.[3] This leads to more intense floods and increased flood risk.[4][5]

Natural types of floods include river flooding, groundwater flooding coastal flooding and urban flooding sometimes known as flash flooding. Tidal flooding may include elements of both river and coastal flooding processes in estuary areas. There is also the intentional flooding of land that would otherwise remain dry. This may take place for agricultural, military, or river-management purposes. For example, agricultural flooding may occur in preparing paddy fields for the growing of semi-aquatic rice in many countries.

Flooding may occur as an overflow of water from water bodies, such as a river, lake, sea or ocean. In these cases, the water overtops or breaks levees, resulting in some of that water escaping its usual boundaries.[6] Flooding may also occur due to an accumulation of rainwater on saturated ground. This is called an *areal flood*. The size of a lake or other body of water naturally varies with seasonal changes in precipitation and snow melt. Those changes in size are however not considered a flood unless they flood property or drown domestic animals.

Floods can also occur in rivers when the flow rate exceeds the capacity of the river channel, particularly at bends or meanders in the waterway. Floods often cause damage to homes and businesses if these buildings are in the natural flood plains of rivers. People could avoid riverine flood damage by moving away from rivers. However, people in many countries have traditionally lived and worked by rivers because the land is usually flat and fertile. Also, the rivers provide easy travel and access to commerce and industry.

Flooding can damage property and also lead to secondary impacts. These include in the short term an increased spread of waterborne diseases and vector-bourne disesases, for example those diseases transmitted by mosquitos. Flooding can also lead to long-term displacement of residents.[7] Floods are an area of study of hydrology and hydraulic engineering.

Types

Floods can happen on flat or low-lying areas when water is supplied by rainfall or snowmelt more rapidly than it can either infiltrate or run off. The excess accumulates in place, sometimes to hazardous depths. Surface soil can become saturated, which effectively stops infiltration, where the water table is shallow, such as a floodplain, or from intense rain from one or a series of storms. Infiltration also is slow to negligible through frozen ground, rock, concrete, paving, or roofs. Areal flooding begins in flat areas like floodplains and in local depressions not connected to a stream channel, because the velocity of overland flow depends on the surface slope. Endorheic basins may experience areal flooding during periods when precipitation exceeds evaporation.[8]

River flooding

Floods occur in all types of river and stream channels, from the smallest ephemeral streams in humid zones to normally-dry channels in arid climates to the world's largest rivers. When overland flow occurs on tilled fields, it can result in a muddy flood where sediments are picked up by run off and carried as suspended matter or bed load. Localized flooding may be caused or exacerbated by drainage obstructions such as landslides, ice, debris, or beaver dams.

Slow-rising floods most commonly occur in large rivers with large catchment areas. The increase in flow may be the result of sustained rainfall, rapid snow melt, monsoons, or tropical cyclones. However, large rivers may have rapid flooding events in areas with dry climates, since they may have large basins but small river channels, and rainfall can be very intense in smaller areas of those basins.

In extremely flat areas, such as the Red River Valley of the North in Minnesota, North Dakota, and Manitoba, a type of hybrid river/areal flooding can occur, known locally as "overland flooding". This is different from "overland flow" defined as "surface runoff". The Red River Valley is a former glacial lakebed, created by Lake Agassiz, and over a length of 550 mi (890 km), the river course drops only 236 ft (72 m), for an average slope of about 5 inches per mile (or 8.2 cm per kilometer).[9] In this very large area, spring snowmelt happens at different rates in different places, and if winter snowfall was heavy, a fast snowmelt can push water out of the banks of a tributary river so that it moves overland, to a point further downstream in the river or completely to another streambed. Overland flooding can be devastating because it is unpredictable, it can occur very suddenly with surprising speed, and in such flat land it can run for miles. It is these qualities that set it apart from simple "overland flow".

Rapid flooding events, including flash floods, more often occur on smaller rivers, rivers with steep valleys, rivers that flow for much of their length over impermeable terrain, or normally-dry channels. The cause may be localized convective precipitation (intense thunderstorms) or sudden release from an upstream impoundment created behind a dam, landslide, or glacier. In one instance, a flash flood killed eight people enjoying the water on a Sunday afternoon at a popular waterfall in a narrow canyon.[*citation needed*] Without any observed rainfall, the flow rate increased from about 50 to 1,500 cubic feet per second (1.4 to 42 m^3/s) in just one minute.[10] Two larger floods occurred at the same site within a week, but no one was at the waterfall on those days. The deadly flood resulted from a thunderstorm over part of the drainage basin, where steep, bare rock slopes are common and the thin soil was already saturated.

Flash floods are the most common flood type in normally-dry channels in arid zones, known as arroyos in the southwest United States and many other names elsewhere. In that setting, the first flood water to arrive is depleted as it wets the sandy stream bed. The leading edge of the flood thus advances more slowly than later and higher flows. As a result, the rising limb of the hydrograph becomes ever quicker as the flood moves downstream, until the flow rate is so great that the depletion by wetting soil becomes insignificant.

Coastal flooding

Coastal areas may be flooded by storm surges combining with high tides and large wave events at sea, resulting in waves over-topping flood defenses or in severe cases by tsunami or tropical cyclones. A storm surge, from either a tropical cyclone or an extratropical cyclone, falls within this category. A storm surge is "an additional rise of water generated by a storm, over and above the predicted astronomical tides".[11] Due to the effects of climate change (e.g. sea level rise and an increase in extreme weather events) and an increase in the population living in coastal areas, the damage caused by coastal flood events has intensified and more people are being affected.[12]

Flooding in estuaries is commonly caused by a combination of storm surges caused by winds and low barometric pressure and large waves meeting high upstream river flows.

Urban flooding

Urban flooding is the inundation of land or property in cities or other built environment, caused by rainfall or coastal storm surges overwhelming the capacity of drainage systems, such as storm sewers. Urban flooding can occur regardless of whether or not affected communities are located within designated floodplains or near any body of water.[13] It is triggered for example by an overflow of rivers and lakes, flash flooding or snowmelt. During the flood, stormwater or water released from damaged water mains may accumulate on property and in public rights-of-way. It can seep through building walls and floors, or backup into buildings through sewer pipes, cellars, toilets and sinks. There are several types of urban flooding, each with a different cause. City planners distinguish *pluvial flooding* (flooding caused by heavy rain), *fluvial flooding* (caused by a nearby river overflowing its banks), or *coastal flooding* (often caused by storm

surges). Urban flooding is a hazard to both the population and infrastructure. Some well known disaster events include the inundations of Nîmes (France) in 1998 and Vaison-la-Romaine (France) in 1992, the flooding of New Orleans (United States) in 2005, and the flooding in Rockhampton, Bundaberg, Brisbane during the 2010–2011 Queensland floods in Australia, the 2022 eastern Australia floods, and more recently the 2024 Rio Grande do Sul floods in Brazil.

Intentional floods

The intentional flooding of land that would otherwise remain dry may take place for agricultural, military or river-management purposes. This is a form of hydraulic engineering. Agricultural flooding may occur in preparing paddy fields for the growing of semi-aquatic rice in many countries.

Flooding for river management may occur in the form of diverting flood waters in a river at flood stage upstream from areas that are considered more valuable than the areas that are sacrificed in this way. This may be done *ad hoc*,[14] or permanently, as in the so-called *overlaten* (literally "let-overs"), an intentionally lowered segment in

Military inundation creates an obstacle in the field that is intended to impede the movement of the enemy.[15] This may be done both for offensive and defensive purposes. Furthermore, in so far as the methods used are a form of hydraulic engineering, it may be useful to differentiate between controlled inundations and uncontrolled ones. Examples for controlled inundations include those in the Netherlands under the Dutch Republic and its successor states in that area[16][17] and exemplified in the two Hollandic Water Lines, the Stelling van Amsterdam, the Frisian Water Line, the IJssel Line, the Peel-Raam Line, and the Grebbe line in that country.

To count as *controlled*, a military inundation has to take the interests of the civilian population into account, by allowing them a timely evacuation, by making the inundation *reversible*, and by making an attempt to minimize the adverse ecological impact of the inundation. That impact may also be adverse in a hydrogeological sense if the inundation lasts a long time.[18]

Floods are caused by many factors or a combination of any of these generally prolonged heavy rainfall (locally concentrated or throughout a catchment area), highly accelerated snowmelt, severe winds over water, unusual high tides, tsunamis, or failure of dams, levees, retention ponds, or other structures that retained the water. Flooding can be exacerbated by increased amounts of impervious surface or by other natural hazards such as wildfires, which reduce the supply of vegetation that can absorb rainfall.

During times of rain, some of the water is retained in ponds or soil, some is absorbed by grass and vegetation, some evaporates, and the rest travels over the land as surface runoff. Floods occur when ponds, lakes, riverbeds, soil, and vegetation cannot absorb all the water.

This has been exacerbated by human activities such as draining wetlands that naturally store large amounts of water and building paved surfaces that do not absorb any water.[21] Water then runs off the land in quantities that cannot be carried within stream channels or retained in natural ponds, lakes, and human-made reservoirs. About 30 percent of all precipitation becomes runoff[22] and that amount might be increased by water from melting snow.

River flooding is often caused by heavy rain, sometimes increased by melting snow. A flood that rises rapidly, with little or no warning, is called a flash flood. Flash floods usually result from intense rainfall over a relatively small area, or if the area was already saturated from previous precipitation.

Periodic floods occur on many rivers, forming a surrounding region known as the flood plain. Even when rainfall is relatively light, the shorelines of lakes and bays can be flooded by severe winds—such as during hurricanes—that blow water into the shore areas.

Upslope factors

The amount, location, and timing of water reaching a drainage channel from natural precipitation and controlled or uncontrolled reservoir releases determines the flow at downstream locations. Some precipitation evaporates, some slowly percolates through soil, some may be temporarily sequestered as snow or ice, and some may produce rapid runoff from surfaces including rock, pavement, roofs, and saturated or frozen ground. The fraction of incident precipitation promptly reaching a drainage channel has been observed from nil for light rain on dry, level ground to as high as 170 percent for warm rain on accumulated snow.[23]

Most precipitation records are based on a measured depth of water received within a fixed time interval. *Frequency* of a precipitation threshold of interest may be determined from the number of measurements exceeding that threshold value within the total time period for which observations are available. Individual data points are converted to *intensity* by dividing each measured depth by the period of time between observations. This intensity will be less than the actual peak intensity if the *duration* of the rainfall event was less than the fixed time interval for which measurements are reported. Convective precipitation events (thunderstorms) tend to produce shorter duration storm events than orographic precipitation. Duration, intensity, and frequency of rainfall events are important to flood prediction. Short duration precipitation is more significant to flooding within small drainage basins.[24]

The most important upslope factor in determining flood magnitude is the land area of the watershed upstream of the area of interest. Rainfall intensity is the second most important factor for watersheds of less than approximately 30 square miles or 80 square kilometres. The main channel slope is the second most important factor for larger watersheds. Channel slope and rainfall intensity become the third most important factors for small and large watersheds, respectively.[25]

Time of Concentration is the time required for runoff from the most distant point of the upstream drainage area to reach the point of the drainage channel controlling flooding of the area of interest. The time of concentration defines the critical duration of peak rainfall for the area of interest.[26] The critical duration of intense rainfall might be only a few minutes for roof and parking lot drainage structures, while cumulative rainfall over several days would be critical for river basins.

Downslope factors

Water flowing downhill ultimately encounters downstream conditions slowing movement. The final limitation in coastal flooding lands is often the ocean or some coastal flooding bars which form natural lakes. In flooding low lands, elevation changes such as tidal fluctuations are significant determinants of coastal and estuarine flooding. Less predictable events like tsunamis and storm surges may also cause elevation changes in large bodies of water. Elevation of flowing water is controlled by the geometry of the flow channel and, especially, by depth of channel, speed of flow and amount of sediments in it[25] Flow channel restrictions like bridges and canyons tend to control water elevation above the restriction. The actual control point for any given reach of the drainage may change with changing water elevation, so a closer point may control for lower water levels until a more distant point controls at higher water levels.

Effective flood channel geometry may be changed by growth of vegetation, accumulation of ice or debris, or construction of bridges, buildings, or levees within the flood channel.

Climate change

Due to an increase in heavy rainfall events, floods are likely to become more severe when they do occur.[29]:1155 The interactions between rainfall and flooding are complex. There are some regions in which flooding is expected to become rarer. This depends on several factors. These include changes in rain and snowmelt, but also soil moisture.[29]:1156 Climate change leaves soils drier in some areas, so they may absorb rainfall more quickly. This leads to less flooding. Dry soils can also become harder. In this case heavy rainfall runs off into rivers and lakes. This increases risks of flooding.[29]:1155

Coincidence

Extreme flood events often result from coincidence such as unusually intense, warm rainfall melting heavy snow pack, producing channel obstructions from floating ice, and releasing small impoundments like beaver dams.[30] Coincident events may cause extensive flooding to be more frequent than anticipated from simplistic statistical prediction models considering only precipitation runoff flowing within unobstructed drainage channels.[31] Debris modification of channel geometry is common when heavy flows move uprooted woody vegetation and flood-damaged structures and vehicles, including boats and railway equipment. Recent field measurements during the 2010–11 Queensland floods showed that any criterion solely based upon the flow velocity, water depth or specific momentum cannot account for the hazards caused by velocity and water depth fluctuations.[32] These considerations ignore further the risks associated with large debris entrained by the flow motion.[33]

Negative impacts

Floods can also be a huge destructive power. When water flows, it has the ability to demolish all kinds of buildings and objects, such as bridges, structures, houses, trees, and cars. Economical, social and natural environmental damages are common factors that are impacted by flooding events and the impacts that flooding has on these areas can be catastrophic.[34]

There have been numerous flood incidents around the world which have caused devastating damage to infrastructure, the natural environment and human life.[34] Flood risks can be defined as the risk that floods pose to individuals, property and the natural landscape based on specific hazards and vulnerability. The extent of flood risks can impact the types of mitigation strategies required and implemented.[35]

Floods can have devastating impacts to human societies. Flooding events worldwide are increasing in frequency and severity, leading to increasing costs to societies.[34] A large amount of the world's population lives in close proximity to major coastlines,[36] while many major cities and agricultural areas are located near floodplains.[37] There is significant risk for increased coastal and fluvial flooding due to changing climatic conditions.[38]

Catastrophic riverine flooding can result from major infrastructure failures, often the collapse of a dam. It can also be caused by drainage channel modification from a landslide, earthquake or volcanic eruption. Examples include outburst floods and lahars. Tsunamis can cause catastrophic coastal flooding, most commonly resulting from undersea earthquakes.

Economic impacts

The primary effects of flooding include loss of life and damage to buildings and other structures, including bridges, sewerage systems, roadways, and canals. The economic impacts caused by flooding can be severe.[37]

Every year flooding causes countries billions of dollars worth of damage that threatens the livelihood of individuals.[39] As a result, there is also significant socio-economic threats to vulnerable populations around the world from flooding.[39] For example, in Bangladesh in 2007, a flood was responsible for the destruction of more than one million houses. And yearly in the United States, floods cause over $7 billion in damage.[40]

Flood waters typically inundate farm land, making the land unworkable and preventing crops from being planted or harvested, which can lead to shortages of food both for humans and farm animals. Entire harvests for a country can be lost in extreme flood circumstances. Some tree species may not survive prolonged flooding of their root systems.[41]

Flooding in areas where people live also has significant economic implications for affected neighborhoods. In the United States, industry experts estimate that wet basements can lower property values by 10–25 percent and are cited among the top reasons for not purchasing a home.[42] According to the U.S. Federal Emergency Management Agency (FEMA), almost 40 percent of small businesses never reopen their doors

following a flooding disaster.[43] In the United States, insurance is available against flood damage to both homes and businesses.[44]

Economic hardship due to a temporary decline in tourism, rebuilding costs, or food shortages leading to price increases is a common after-effect of severe flooding. The impact on those affected may cause psychological damage to those affected, in particular where deaths, serious injuries and loss of property occur.

Health impacts

Fatalities connected directly to floods are usually caused by drowning; the waters in a flood are very deep and have strong currents.[45] Deaths do not just occur from drowning, deaths are connected with dehydration, heat stroke, heart attack and any other illness that needs medical supplies that cannot be delivered.[45]

Injuries can lead to an excessive amount of morbidity when a flood occurs. Injuries are not isolated to just those who were directly in the flood, rescue teams and even people delivering supplies can sustain an injury. Injuries can occur anytime during the flood process; before, during and after.[45] During floods accidents occur with falling debris or any of the many fast moving objects in the water. After the flood rescue attempts are where large numbers injuries can occur.[45]

Communicable diseases are increased due to many pathogens and bacteria that are being transported by the water.There are many waterborne diseases such as cholera, hepatitis A, hepatitis E and diarrheal diseases, to mention a few. Gastrointestinal disease and diarrheal diseases are very common due to a lack of clean water during a flood. Most of clean water supplies are contaminated when flooding occurs. Hepatitis A and E are common because of the lack of sanitation in the water and in living quarters depending on where the flood is and how prepared the community is for a flood.[45]

When floods hit, people lose nearly all their crops, livestock, and food reserves and face starvation.[46]

Floods also frequently damage power transmission and sometimes power generation, which then has knock-on effects caused by the loss of power. This includes loss of drinking water treatment and water supply, which may result in loss of drinking water or severe water contamination. It may also cause the loss of sewage disposal facilities. Lack of clean water combined with human sewage in the flood waters raises the risk of waterborne diseases, which can include typhoid, giardia, cryptosporidium, cholera and many other diseases depending upon the location of the flood.

Damage to roads and transport infrastructure may make it difficult to mobilize aid to those affected or to provide emergency health treatment.

Flooding can cause chronically wet houses, leading to the growth of indoor mold and resulting in adverse health effects, particularly respiratory symptoms.[47] Respiratory diseases are a common after the disaster has occurred. This depends on the amount of water damage and mold that grows after an incident. Research suggests that there will be an increase of 30–50% in adverse respiratory health outcomes caused by dampness and mold exposure for those living in coastal and wetland areas. Fungal contamination in homes is associated with increased allergic rhinitis and asthma.[48] Vector borne diseases increase as well due to the increase in still water after the floods have settled. The diseases that are vector borne are malaria, dengue, West Nile, and yellow fever.[45] Floods have a huge impact on victims' psychosocial integrity. People suffer from a wide variety of losses and stress. One of the most treated illness in long-term health problems are depression caused by the flood and all the tragedy that flows with one.[45]

Loss of life

Main article: List of deadliest floods

Below is a list of the deadliest floods worldwide, showing events with death tolls at or above 100,000 individuals.

Death toll	Event	Location	Year
2,500,000–3,700,000[49]	1931 China floods	China	1931
900,000–2,000,000	1887 Yellow River flood	China	1887
500,000–700,000	1938 Yellow River flood	China	1938
231,000	Banqiao Dam failure, result of Typhoon Nina. Approximately 86,000 people died from flooding and another 145,000 died during subsequent disease.	China	1975
230,000	2004 Indian Ocean tsunami	Indonesia	2004
145,000	1935 Yangtze river flood	China	1935
100,000+	St. Felix's flood, storm surge	Netherlands	1530
100,000	Hanoi and Red River Delta flood	North Vietnam	1971
100,000	1911 Yangtze river flood	China	1911

Positive impacts (benefits)

Floods (in particular more frequent or smaller floods) can also bring many benefits, such as recharging ground water, making soil more fertile and increasing nutrients in some soils. Flood waters provide much needed water resources in arid and semi-arid regions where precipitation can be very unevenly distributed throughout the year and kills pests in the farming land. Freshwater floods particularly play an important role in maintaining ecosystems in river corridors and are a key factor in maintaining floodplain biodiversity.[50] Flooding can spread nutrients to lakes and rivers, which can lead to increased biomass and improved fisheries for a few years.

For some fish species, an inundated floodplain may form a highly suitable location for spawning with few predators and enhanced levels of nutrients or food.[51] Fish, such as the weather fish, make use of floods in order to reach new habitats. Bird populations may also profit from the boost in food production caused by flooding.[52]

Flooding can bring benefits, such as making the soil more fertile and providing it with more nutrients. For this reason, periodic flooding was essential to the well-being of ancient communities along the Tigris-Euphrates Rivers, the Nile River, the Indus River, the Ganges and the Yellow River among others.

The viability of hydropower, a renewable source of energy, is also higher in flood prone regions.

Protections against floods and associated hazards

Flood control

In many countries around the world, waterways prone to floods are often carefully managed. Defenses such as detention basins, levees,[53] bunds, reservoirs, and weirs are used to prevent waterways from overflowing

their banks. When these defenses fail, emergency measures such as sandbags or portable inflatable tubes are often used to try to stem flooding. Coastal flooding has been addressed in portions of Europe and the Americas with coastal defenses, such as sea walls, beach nourishment, and barrier islands.

In the riparian zone near rivers and streams, erosion control measures can be taken to try to slow down or reverse the natural forces that cause many waterways to meander over long periods of time. Flood controls, such as dams, can be built and maintained over time to try to reduce the occurrence and severity of floods as well. In the United States, the U.S. Army Corps of Engineers maintains a network of such flood control dams.

In areas prone to urban flooding, one solution is the repair and expansion of human-made sewer systems and stormwater infrastructure. Another strategy is to reduce impervious surfaces in streets, parking lots and buildings through natural drainage channels, porous paving, and wetlands (collectively known as green infrastructure or sustainable urban drainage systems (SUDS)). Areas identified as flood-prone can be converted into parks and playgrounds that can tolerate occasional flooding. Ordinances can be adopted to require developers to retain stormwater on site and require buildings to be elevated, protected by floodwalls and levees, or designed to withstand temporary inundation. Property owners can also invest in solutions themselves, such as re-landscaping their property to take the flow of water away from their building and installing rain barrels, sump pumps, and check valves.

In some areas, the presence of certain species (such as beavers) can be beneficial for flood control reasons. Beavers build and maintain beaver dams which will reduce the height of flood waves moving down the river (during periods of heavy rains), and will reduce or eliminate damage to human structures,[54][55] at the cost of minor flooding near the dams (often on farmland). Besides this, they also boost wildlife populations and filter pollutants (manure, fertilisers, slurry).[54] UK environment minister Rebecca Pow stated that in the future the beavers could be considered a "public good" and landowners would be paid to have them on their land.[56]

Flood risk management

Flood management describes methods used to reduce or prevent the detrimental effects of flood waters. Flooding can be caused by a mix of both natural processes, such as extreme weather upstream, and human changes to waterbodies and runoff.

Natural flooding has many beneficial environmental effects [57] and is a seasonal occurrence in many countries where floods help replenish soil fertility, restore wet-lands and promote bio-diversity.[58] It is only where floods interact with housing, industry and farming that flood management is indicated and in such cases environmentally helpful solutions may provide solutions,.[59]

Flood management methods can be either of the *structural* type (i.e. flood control) and of the *non-structural* type. Structural methods hold back floodwaters physically, while non-structural methods do not. Building hard infrastructure to prevent flooding, such as flood walls, is effective at managing flooding. However, it is best practice within landscape engineering to rely more on soft infrastructure and natural systems, such as marshes and flood plains, for handling the increase in water.

As climate change has led to increased flood risk an intensity, flood management is an important part of climate change adaptation and climate resilience.[60][61] For example, to prevent or manage coastal flooding, coastal management practices have to handle natural processes like tides but also sea level rise due to climate change. The prevention and mitigation of flooding can be studied on three levels: on individual properties, small communities, and whole towns or cities.

In the United States, the National Weather Service gives out the advice "Turn Around, Don't Drown" for floods; that is, it recommends that people get out of the area of a flood, rather than trying to cross it. At the most basic level, the best defense against floods is to seek higher ground for high-value uses while balancing the foreseeable risks with the benefits of occupying flood hazard zones.[62]:22–23 Critical community-safety facilities, such as hospitals, emergency-operations centers, and police, fire, and rescue services, should be

built in areas least at risk of flooding. Structures, such as bridges, that must unavoidably be in flood hazard areas should be designed to withstand flooding. Areas most at risk for flooding could be put to valuable uses that could be abandoned temporarily as people retreat to safer areas when a flood is imminent.

Planning for flood safety involves many aspects of analysis and engineering, including:

- observation of previous and present flood heights and inundated areas,
- statistical, hydrologic, and hydraulic model analyses,
- mapping inundated areas and flood heights for future flood scenarios,
- long-term land use planning and regulation,
- engineering design and construction of structures to control or withstand flooding,
- intermediate-term monitoring, forecasting, and emergency-response planning, and
- short-term monitoring, warning, and response operations.

Each topic presents distinct yet related questions with varying scope and scale in time, space, and the people involved. Attempts to understand and manage the mechanisms at work in floodplains have been made for at least six millennia.[63][*page needed*]

In the United States, the Association of State Floodplain Managers works to promote education, policies, and activities that mitigate current and future losses, costs, and human suffering caused by flooding and to protect the natural and beneficial functions of floodplains – all without causing adverse impacts.[64] A portfolio of best practice examples for disaster mitigation in the United States is available from the Federal Emergency Management Agency.[65]

Flood clean-up safety

Clean-up activities following floods often pose hazards to workers and volunteers involved in the effort. Potential dangers include electrical hazards, carbon monoxide exposure, musculoskeletal hazards, heat or cold stress, motor vehicle-related dangers, fire, drowning, and exposure to hazardous materials. Because flooded disaster sites are unstable, clean-up workers might encounter sharp jagged debris, biological hazards in the flood water, exposed electrical lines, blood or other body fluids, and animal and human remains. In planning for and reacting to flood disasters, managers provide workers with hard hats, goggles, heavy work gloves, life jackets, and watertight boots with steel toes and insoles.[66]

Flood predictions

A series of annual maximum flow rates in a stream reach can be analyzed statistically to estimate the 100-year flood and floods of other recurrence intervals there. Similar estimates from many sites in a hydrologically similar region can be related to measurable characteristics of each drainage basin to allow indirect estimation of flood recurrence intervals for stream reaches without sufficient data for direct analysis.

Physical process models of channel reaches are generally well understood and will calculate the depth and area of inundation for given channel conditions and a specified flow rate, such as for use in floodplain mapping and flood insurance. Conversely, given the observed inundation area of a recent flood and the channel conditions, a model can calculate the flow rate. Applied to various potential channel configurations and flow rates, a reach model can contribute to selecting an optimum design for a modified channel. Various reach models are available as of 2015, either 1D models (flood levels measured in the channel) or 2D models (variable flood depths measured across the extent of a floodplain). HEC-RAS,[67] the Hydraulic Engineering Center model, is among the most popular software, if only because it is available free of charge. Other models such as TUFLOW[68] combine 1D and 2D components to derive flood depths across both river channels and the entire floodplain.

Physical process models of complete drainage basins are even more complex. Although many processes are well understood at a point or for a small area, others are poorly understood at all scales, and process

interactions under normal or extreme climatic conditions may be unknown. Basin models typically combine land-surface process components (to estimate how much rainfall or snowmelt reaches a channel) with a series of reach models. For example, a basin model can calculate the runoff hydrograph that might result from a 100-year storm, although the recurrence interval of a storm is rarely equal to that of the associated flood. Basin models are commonly used in flood forecasting and warning, as well as in analysis of the effects of land use change and climate change.

In the United States, an integrated approach to real-time hydrologic computer modelling uses observed data from the U.S. Geological Survey (USGS),[69] various cooperative observing networks,[70] various automated weather sensors, the NOAA National Operational Hydrologic Remote Sensing Center (NOHRSC),[71] various hydroelectric companies, etc. combined with quantitative precipitation forecasts (QPF) of expected rainfall and/or snow melt to generate daily or as-needed hydrologic forecasts.[72] The NWS also cooperates with Environment Canada on hydrologic forecasts that affect both the US and Canada, like in the area of the Saint Lawrence Seaway.

The Global Flood Monitoring System, "GFMS", a computer tool which maps flood conditions worldwide, is available online.[73] Users anywhere in the world can use GFMS to determine when floods may occur in their area. GFMS uses precipitation data from NASA's Earth observing satellites and the Global Precipitation Measurement satellite, "GPM". Rainfall data from GPM is combined with a land surface model that incorporates vegetation cover, soil type, and terrain to determine how much water is soaking into the ground, and how much water is flowing into streamflow.

Users can view statistics for rainfall, streamflow, water depth, and flooding every 3 hours, at each 12-kilometer gridpoint on a global map. Forecasts for these parameters are 5 days into the future. Users can zoom in to see inundation maps (areas estimated to be covered with water) in 1-kilometer resolution.[74]

Flood forecasts and warnings

Anticipating floods before they occur allows for precautions to be taken and people to be warned[75] so that they can be prepared in advance for flooding conditions. For example, farmers can remove animals from low-lying areas and utility services can put in place emergency provisions to re-route services if needed. Emergency services can also make provisions to have enough resources available ahead of time to respond to emergencies as they occur. People can evacuate areas to be flooded.

In order to make the most accurate flood forecasts for waterways, it is best to have a long time-series of historical data that relates stream flows to measured past rainfall events.[76] Coupling this historical information with real-time knowledge about volumetric capacity in catchment areas, such as spare capacity in reservoirs, ground-water levels, and the degree of saturation of area aquifers is also needed in order to make the most accurate flood forecasts.

Radar estimates of rainfall and general weather forecasting techniques are also important components of good flood forecasting. In areas where good quality data is available, the intensity and height of a flood can be predicted with fairly good accuracy and plenty of lead time. The output of a flood forecast is typically a maximum expected water level and the likely time of its arrival at key locations along a waterway,[72] and it also may allow for the computation of the likely statistical return period of a flood. In many developed countries, urban areas at risk of flooding are protected against a 100-year flood – that is a flood that has a probability of around 63% of occurring in any 100-year period of time.

According to the U.S. National Weather Service (NWS) Northeast River Forecast Center (RFC) in Taunton, Massachusetts, a rule of thumb for flood forecasting in urban areas is that it takes at least 1 inch (25 mm) of rainfall in around an hour's time in order to start significant ponding of water on impermeable surfaces. Many NWS RFCs routinely issue Flash Flood Guidance and Headwater Guidance, which indicate the general amount of rainfall that would need to fall in a short period of time in order to cause flash flooding or flooding on larger water basins.[77]

Flooding is a temporary overflow of water onto land that is normally dry. Floods are the most common disaster in the United States. Failing to evacuate flooded areas or entering flood waters can lead to injury or death.

Floods may:

- Result from rain, snow, coastal storms, storm surges and overflows of dams and other water systems.
- Develop slowly or quickly. Flash floods can come with no warning.
- Cause outages, disrupt transportation, damage buildings and create landslides.

If you are under a flood warning:

- Find safe shelter right away.
- Do not walk, swim or drive through flood waters**. Turn Around, Don't Drown!**
- Remember, just six inches of moving water can knock you down, and one foot of moving water can sweep your vehicle away.
- Stay off bridges over fast-moving water.
- Depending on the type of flooding:
    - Evacuate if told to do so.
    - Move to higher ground or a higher floor.
    - Stay where you are.

Preparing for a Flood

Know Your Risk for Floods
Visit FEMA's Flood Map Service Center to know types of flood risk in your area. Sign up for your community's warning system. The Emergency Alert System (EAS) and National Oceanic and Atmospheric Administration (NOAA) Weather Radio also provide emergency alerts.

Purchase Flood Insurance
Purchase or renew a flood insurance policy. Homeowner's insurance policies do not cover flooding. It typically takes up to 30 days for a policy to go into effect so the time to buy is well before a disaster. Get flood coverage under the National Flood Insurance Program (NFIP).

Preparing for a Flood
Make a plan for your household, including your pets, so that you and your family know what to do, where to go, and what you will need to protect yourselves from flooding. Learn and practice evacuation routes, shelter plans, and flash flood response. Gather supplies, including non-perishable foods, cleaning supplies, and water for several days, in case you must leave immediately or if services are cut off in your area.

In Case of Emergency
Keep important documents in a waterproof container. Create password-protected digital copies. Protect your property. Move valuables to higher levels. Declutter drains and gutters. Install check valves. Consider a sump pump with a battery.

Staying Safe During a Flood

- Evacuate immediately, if told to evacuate. Never drive around barricades. Local responders use them to safely direct traffic out of flooded areas.
- Contact your healthcare provider If you are sick and need medical attention. Wait for further care instructions and shelter in place, if possible. If you are experiencing a medical emergency, call 9-1-1.
- Listen to EAS, NOAA Weather Radio or local alerting systems for current emergency information and instructions regarding flooding.
- Do not walk, swim or drive through flood waters. **Turn Around. Don't Drown!**
- Stay off bridges over fast-moving water. Fast-moving water can wash bridges away without warning.
- Stay inside your car if it is trapped in rapidly moving water. Get on the roof if water is rising inside the car.
- Get to the highest level if trapped in a building. Only get on the roof if necessary and once there signal for help. Do not climb into a closed attic to avoid getting trapped by rising floodwater.

Staying Safe After a Flood

- Pay attention to authorities for information and instructions. Return home only when authorities say it is safe.
- Avoid driving except in emergencies.
- Wear heavy work gloves, protective clothing and boots during clean up and use appropriate face coverings or masks if cleaning mold or other debris.
- People with asthma and other lung conditions and/or immune suppression should not enter buildings with indoor water leaks or mold growth that can be seen or smelled. Children should not take part in disaster cleanup work.
- Be aware that snakes and other animals may be in your house.
- Be aware of the risk of electrocution. Do not touch electrical equipment if it is wet or if you are standing in water. Turn off the electricity to prevent electric shock if it is safe to do so.
- Avoid wading in floodwater, which can be contaminated and contain dangerous debris. Underground or downed power lines can also electrically charge the water.
- Use a generator or other gasoline-powered machinery ONLY outdoors and away from windows.

Floods can be caused by a number of factors, including:

- Heavy rainfall: When the amount of rain exceeds the drainage system's capacity, it can lead to flooding.
- Snowmelt: Rapid snowmelt can cause flooding, particularly in the spring when warm air can cause sudden snowmelt.
- River overflow: Rivers can overflow their banks and cause flooding.
- Deforestation: Trees help to hold soil and water, preventing flooding. When trees are cut down, more water runs towards rivers during heavy rainfall.
- Dam failure: Dams can fail when rainfall is intense and water levels rise. When a dam breaks, it can unleash a large amount of water on downstream areas.

- Coastal flooding: Heavy rainfall can cause waterlogging conditions when rainwater can't discharge through nearby rivers. Coastal inundation can occur when coastal areas are submerged by seawater due to sea-level rise, tidal surges, or cyclones.
- Debris in the river: Debris in the river can block the flow of water and cause flooding

Floods

India is highly vulnerable to floods. Out of the total geographical area of 329 million hectares (mha), more than 40 mha is flood prone. Floods are a recurrent phenomenon, which cause huge loss of lives and damage to livelihood systems, property, infrastructure and public utilities. It is a cause for concern that flood related damages show an increasing trend. The average annual flood damage in the last 10 years period from 1996 to 2005 was Rs. 4745 crore as compared to Rs. 1805 crore, the corresponding average for the previous 53 years. This can be attributed to many reasons including a steep increase in population, rapid urbanization growing developmental and economic activities in flood plains coupled with global warming.

An average every year, 75 lakh hectares of land is affected, 1600 lives are lost and the damage caused to crops, houses and public utilities is Rs.1805 crores due to floods. The maximum number of lives (11,316) was lost in the year 1977. The frequency of major floods is more than once in five years.

Floods have also occurred in areas, which were earlier not considered flood prone. An effort has been made in these Guidelines to cover the entire gamut of Flood Management. Eighty per cent of the precipitation takes place in the monsoon months from June to September. The rivers a bring heavy sediment load from catchments. These, coupled with inadequate carrying capacity of rivers are responsible for causing floods, drainage congestion and erosion of river-banks. Cyclones, cyclonic circulations and cloud bursts cause flash floods and lead to huge losses. It is a fact that some of the rivers causing damage in India originate in neighboring countries; adding another complex dimension to the problem. Continuing and large-scale loss of lives and damage to public and private property due to floods indicate that we are still to develop an effective response to floods. NDMA's Executive Summary Guidelines have been prepared to enable the various implementing and stakeholder agencies to effectively address the critical areas for minimising flood damage.

Flood control measures related to human adjustment, perception, and mitigation include land use planning to avoid floodplains, elevating structures, community awareness campaigns, early warning systems, flood insurance, property buyouts in high-risk areas, relocating settlements, building codes that incorporate flood resilience, and actively engaging communities in risk assessment and decision-making processes to understand their perceptions and incorporate local knowledge into mitigation strategies.

Key aspects of human adjustment and perception in flood control:

- Risk perception:

  Understanding how people perceive the flood risk in their area, which can be influenced by past experiences, cultural factors, and access to information.

- Community engagement:

  Actively involving local communities in planning and decision-making processes to ensure their needs and knowledge are considered.

- Awareness campaigns:
  Educating residents about flood risks, preparedness measures, evacuation routes, and appropriate responses during floods.

Mitigation strategies based on human adjustment and perception:

- Land use planning:

  Identifying and regulating development in floodplains, promoting green spaces to absorb water, and implementing zoning regulations.

- Building codes:

  Enforcing construction standards that elevate buildings and incorporate flood-resistant materials

- Property buyouts:

  Purchasing properties in high-risk flood zones to relocate residents to safer areas

- Floodproofing measures:

  Retrofitting existing structures with flood-resistant features like flood barriers and elevated electrical outlets

- Early warning systems:
  Establishing systems to monitor water levels and provide timely alerts to communities

Important considerations:

- Social vulnerability:

  Recognizing that certain populations, like low-income communities, may be disproportionately affected by floods and require targeted support.

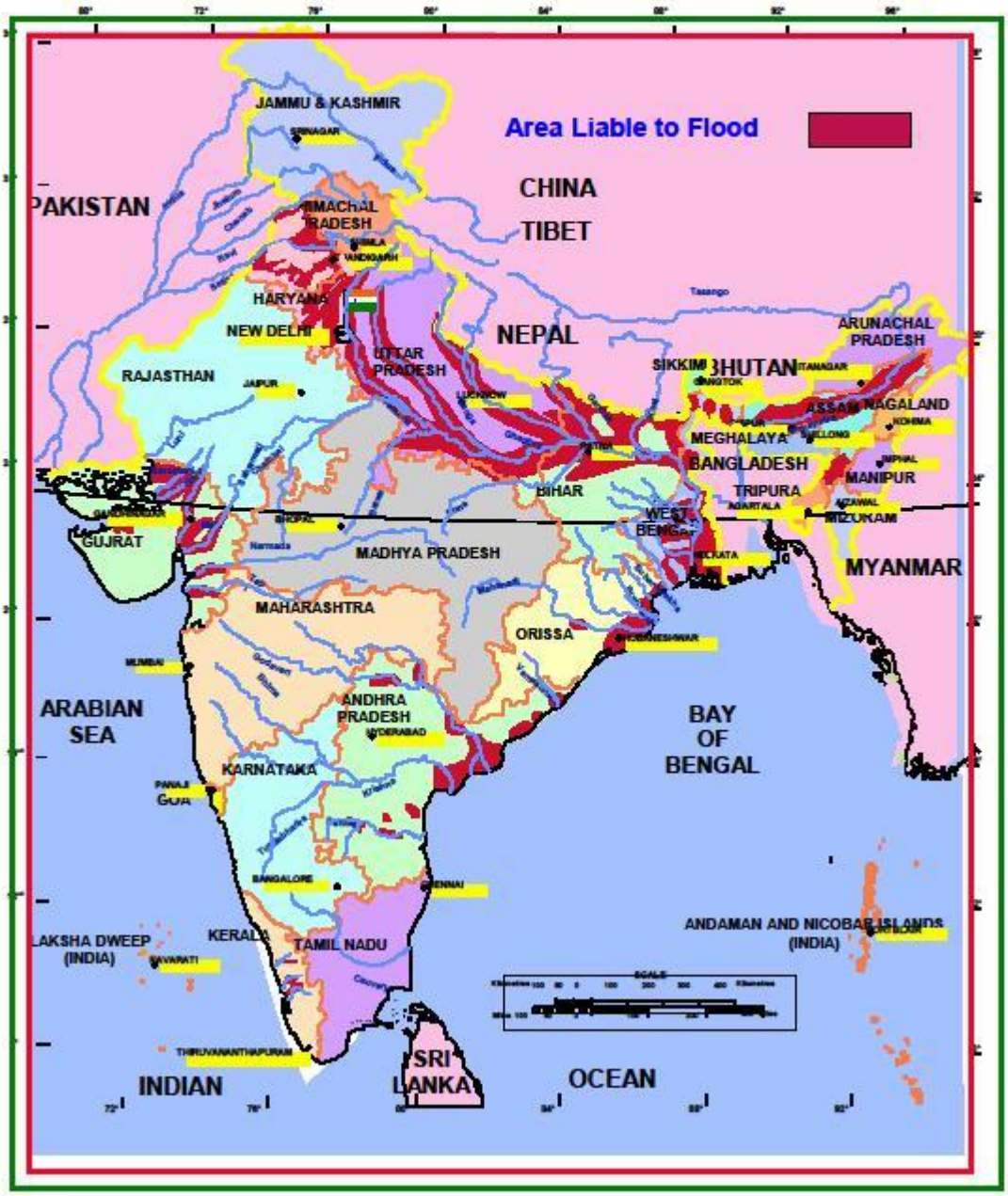
- Cultural sensitivity:

  Respecting local customs and practices when implementing flood control measures.

- Adaptive management:
  Continuously monitoring and adjusting flood control strategies based on changing conditions and community feedback.

INDIA

AREA LIABLE TO FLOODS



Lecture No 7

Drought is a prolonged dry period in the natural climate cycle that can occur anywhere in the world. It is a slow-onset disaster characterized by the lack of precipitation, resulting in a water shortage. Drought can have a serious impact on health, agriculture, economies, energy and the environment.

An estimated 55 million people globally are affected by droughts every year, and they are the most serious hazard to livestock and crops in nearly every part of the world. Drought threatens people's livelihoods, increases the risk of disease and death, and fuels mass migration. Water scarcity impacts 40% of the world's population, and as many as 700 million people are at-risk of being displaced as a result of drought by 2030.

Rising temperatures caused by climate change are making already dry regions drier and wet regions wetter. In dry regions, this means that when temperatures rise, water evaporates more quickly, and thus increases the risk of drought or prolongs periods of drought. Between 80-90% of all documented disasters from natural hazards during the past 10 years have resulted from floods, droughts, tropical cyclones, heat waves and severe storms.

When drought causes water and food shortages there can be many impacts on the health of the affected population, which may increase the risk of disease and death. Drought may have acute and chronic health effects, including:

- malnutrition due to the decreased availability of food, including micronutrient deficiency, such as iron-deficiency anaemia;
- increased risk of infectious diseases, such as cholera, diarrhoea, and pneumonia, due to acute malnutrition, lack of water and sanitation, and displacement;
- psycho-social stress and mental health disorders;
- disruption of local health services due to a lack of water supplies, loss of buying power, migration and/or health workers being forced to leave local areas.

Severe drought can also affect air quality by making wildfires and dust storms more likely, increasing health risk in people already impacted by lung diseases, like asthma or chronic obstructive pulmonary disease (COPD), or with heart disease.

A **drought** is a period of drier-than-normal conditions.[1]:1157 A drought can last for days, months or years. Drought often has large impacts on the ecosystems and agriculture of affected regions, and causes harm to the local economy.[2][3] Annual dry seasons in the tropics significantly increase the chances of a drought developing, with subsequent increased wildfire risks.[4] Heat waves can significantly worsen drought conditions by increasing evapotranspiration.[5] This dries out forests and other vegetation, and increases the amount of fuel for wildfires.[4][6]

Drought is a recurring feature of the climate in most parts of the world, becoming more extreme and less predictable due to climate change, which dendrochronological studies date back to 1900. There are three kinds of drought effects, environmental, economic and social. Environmental effects include the drying of wetlands, more and larger wildfires, loss of biodiversity.

Economic impacts include disruption of water supplies for people, less agricultural productivity and therefore more expensive food production. Another impact is shortages of water for irrigation or hydropower. Social and health costs include the negative effect on the health of people directly exposed to this phenomenon (excessive heat waves), high food costs, stress caused by failed harvests, water scarcity, etc. Prolonged droughts have caused mass migrations and humanitarian crisis.[7][8]

Examples for regions with increased drought risks are the Amazon basin, Australia, the Sahel region and India. For example, in 2005, parts of the Amazon basin experienced the worst drought in 100 years.[9][10] Australia could experience more severe droughts and they could become more frequent in the future, a government-commissioned report said on July 6, 2008.[11] The long Australian Millennial drought broke in 2010. The 2020–2022 Horn of Africa drought has surpassed the horrific drought in 2010–2011 in both duration and severity.[12][13] More than 150 districts in India are drought vulnerable, mostly concentrated in the state of Rajasthan, Gujarat, Madhya Pradesh and its adjoining Chhattisgarh, Uttar Pradesh, northern Karnataka and adjoining Maharashtra of the country.[14]

Throughout history, humans have usually viewed droughts as disasters due to the impact on food availability and the rest of society. People have viewed drought as a natural disaster or as something influenced by human activity, or as a result of supernatural forces.

Definition


Fields outside Benambra, Australia suffering from drought in 2006.

The IPCC Sixth Assessment Report defines a drought simply as "drier than normal conditions".[1]:1157 This means that a drought is "a moisture deficit relative to the average water availability at a given location and season".[1]:1157

According to National Integrated Drought Information System, a multi-agency partnership, drought is generally defined as "a deficiency of precipitation over an extended period of time (usually a season or more), resulting in a water shortage". The National Weather Service office of the NOAA defines drought as "a deficiency of moisture that results in adverse impacts on people, animals, or vegetation over a sizeable area".[15]

Drought is a complex phenomenon − relating to the absence of water − which is difficult to monitor and define.[16] By the early 1980s, over 150 definitions of "drought" had already been published.[17] The range of definitions reflects differences in regions, needs, and disciplinary approaches.

Categories

There are three major categories of drought based on where in the water cycle the moisture deficit occurs: meteorological drought, hydrological drought, and agricultural or ecological drought.[1]:1157 A meteorological drought occurs due to lack of precipitation. A hydrological drought is related to low runoff, streamflow, and reservoir storage. An agricultural or ecological drought is causing plant stress from a combination of evaporation and low soil moisture.[1]:1157 Some organizations add another category: socioeconomic drought occurs when the demand for an economic good exceeds supply as a result of a weather-related shortfall in water supply.[16][17] The socioeconomic drought is a similar concept to water scarcity.

The different categories of droughts have different causes but similar effects:

1. Meteorological drought occurs when there is a prolonged time with less than average precipitation.[18] Meteorological drought usually precedes the other kinds of drought.[19] As a drought persists, the conditions surrounding it gradually worsen and its impact on the local population gradually increases.

2. Hydrological drought is brought about when the water reserves available in sources such as aquifers, lakes and reservoirs fall below a locally significant threshold. Hydrological drought tends to show up more slowly because it involves stored water that is used but not replenished. Like an agricultural drought, this can be triggered by more than just a loss of rainfall. For instance, around 2007 Kazakhstan was awarded a large amount of money by the World Bank to restore water that had been diverted to other nations from the Aral Sea under Soviet rule.[20] Similar circumstances also place their largest lake, Balkhash, at risk of completely drying out.[21]
3. Agricultural or ecological droughts affect crop production or ecosystems in general. This condition can also arise independently from any change in precipitation levels when either increased irrigation or soil conditions and erosion triggered by poorly planned agricultural endeavors cause a shortfall in water available to the crops.

Indices and monitoring

Several indices have been defined to quantify and monitor drought at different spatial and temporal scales. A key property of drought indices is their spatial comparability, and they must be statistically robust.[22] Drought indices include:[22]

- Palmer drought index (sometimes called the Palmer drought severity index (PDSI)): a regional drought index commonly used for monitoring drought events and studying areal extent and severity of drought episodes.[23] The index uses precipitation and temperature data to study moisture supply and demand using a simple water balance model.[23][24][25]
- Keetch-Byram Drought Index: an index that is calculated based on rainfall, air temperature, and other meteorological factors.[26]
- Standardized precipitation index (SPI): It is computed based on precipitation, which makes it a simple and easy-to-apply indicator for monitoring and prediction of droughts in different parts of the world. The World Meteorological Organization recommends this index for identifying and monitoring meteorological droughts in different climates and time periods.[22]
- Standardized Precipitation Evapotranspiration Index (SPEI): a multiscalar drought index based on climatic data. The SPEI accounts also for the role of the increased atmospheric evaporative demand on drought severity.[22] Evaporative demand is particularly dominant during periods of precipitation deficit. The SPEI calculation requires long-term and high-quality precipitation and atmospheric evaporative demand datasets. These can be obtained from ground stations or gridded data based on reanalysis as well as satellite and multi-source datasets.[22]
- Indices related to vegetation: root-zone soil moisture, vegetation condition index (VDI) and vegetation health index (VHI). The VCI and VHI are computed based on vegetation indices such as the normalized difference vegetation index (NDVI) and temperature datasets.[22]
- Deciles index
- Standardized runoff index

High-resolution drought information helps to better assess the spatial and temporal changes and variability in drought duration, severity, and magnitude at a much finer scale. This supports the development of site-specific adaptation measures.[22]

The application of multiple indices using different datasets helps to better manage and monitor droughts than using a single dataset, This is particularly the case in regions of the world where not enough data is available such as Africa and South America. Using a single dataset can be limiting, as it may not capture the full spectrum of drought characteristics and impacts.[22]

Careful monitoring of moisture levels can also help predict increased risk for wildfires.

Causes

 Contraction and desiccation cracks in the dry earth of the Sonoran desert, northwestern Mexico

General precipitation deficiency

See also: Precipitation

Mechanisms of producing precipitation include convective, stratiform,[27] and orographic rainfall.[28] Convective processes involve strong vertical motions that can cause the overturning of the atmosphere in that location within an hour and cause heavy precipitation,[29] while stratiform processes involve weaker upward motions and less intense precipitation over a longer duration.[30]

Precipitation can be divided into three categories, based on whether it falls as liquid water, liquid water that freezes on contact with the surface, or ice.

Droughts occur mainly in areas where normal levels of rainfall are, in themselves, low. If these factors do not support precipitation volumes sufficiently to reach the surface over a sufficient time, the result is a drought. Drought can be triggered by a high level of reflected sunlight and above average prevalence of high pressure systems, winds carrying continental, rather than oceanic air masses, and ridges of high pressure areas aloft can prevent or restrict the developing of thunderstorm activity or rainfall over one certain region. Once a region is within drought, feedback mechanisms such as local arid air,[31] hot conditions which can promote warm core ridging,[32] and minimal evapotranspiration can worsen drought conditions.

Dry season

See also: Dry season

Within the tropics, distinct, wet and dry seasons emerge due to the movement of the Intertropical Convergence Zone or Monsoon trough.[33] The dry season greatly increases drought occurrence,[34] and is characterized by its low humidity, with watering holes and rivers drying up. Because of the lack of these watering holes, many grazing animals are forced to migrate due to the lack of water in search of more fertile lands. Examples of such animals are zebras, elephants, and wildebeest. Because of the lack of water in the plants, bushfires are common.[35] Since water vapor becomes more energetic with increasing temperature, more water vapor is required to increase relative humidity values to 100% at higher temperatures (or to get the temperature to fall to the dew point).[36] Periods of warmth quicken the pace of fruit and vegetable production,[37] increase evaporation and transpiration from plants,[38] and worsen drought conditions.[39]

El Niño–Southern Oscillation (ENSO)

Further information: El Niño–Southern Oscillation

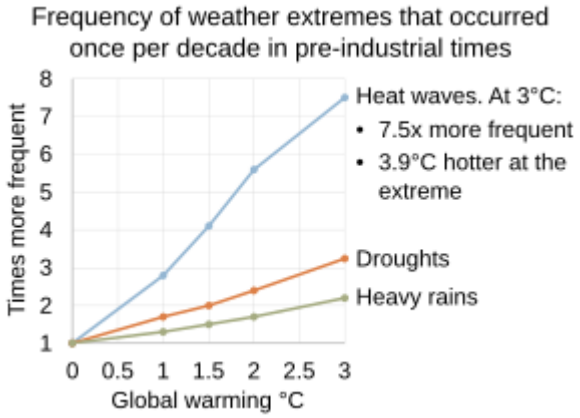
The El Niño–Southern Oscillation (ENSO) phenomenon can sometimes play a significant role in drought. ENSO comprises two patterns of temperature anomalies in the central Pacific Ocean, known as La

Niña and El Niño. La Niña events are generally associated with drier and hotter conditions and further exacerbation of drought in California and the Southwestern United States, and to some extent the U.S. Southeast. Meteorological scientists have observed that La Niñas have become more frequent over time.[40]

Conversely, during El Niño events, drier and hotter weather occurs in parts of the Amazon River Basin, Colombia, and Central America. Winters during the El Niño are warmer and drier than average conditions in the Northwest, northern Midwest, and northern Mideast United States, so those regions experience reduced snowfalls. Conditions are also drier than normal from December to February in south-central Africa, mainly in Zambia, Zimbabwe, Mozambique, and Botswana. Direct effects of El Niño resulting in drier conditions occur in parts of Southeast Asia and Northern Australia, increasing bush fires, worsening haze, and decreasing air quality dramatically. Drier-than-normal conditions are also in general observed in Queensland, inland Victoria, inland New South Wales, and eastern Tasmania from June to August. As warm water spreads from the west Pacific and the Indian Ocean to the east Pacific, it causes extensive drought in the western Pacific. Singapore experienced the driest February in 2014 since records began in 1869, with only 6.3 mm of rain falling in the month and temperatures hitting as high as 35 °C on 26 February. The years 1968 and 2005 had the next driest Februaries, when 8.4 mm of rain fell.[41]

Climate change

See also: Effects of climate change on the water cycle and Effects of climate change on agriculture



There will likely be multiplicative increases in the frequency of extreme weather events compared to the pre-industrial era for heat waves, droughts and heavy precipitation events, for various climate change scenarios.[42]

Globally, the occurrence of droughts has increased as a result of the increase in temperature and atmospheric evaporative demand. In addition, increased climate variability has increased the frequency and severity of drought events. Moreover, the occurrence and impact of droughts are aggravated by anthropogenic activities such as land use change and water management and demand.[22]

The IPCC Sixth Assessment Report also pointed out that "Warming over land drives an increase in atmospheric evaporative demand and in the severity of drought events"[43]:1057 and "Increased atmospheric evaporative demand increases plant water stress, leading to agricultural and ecological drought".[44]:578

There is a rise of compound warm-season droughts in Europe that are concurrent with an increase in potential evapotranspiration.[45]

This section is an excerpt from Effects of climate change § Droughts.[edit]

 A dry lakebed in California. In 2022, the state was experiencing its most serious drought in 1,200 years, worsened by climate change.[46]Climate change affects many factors associated with droughts. These include how much rain falls and how fast the rain evaporates again. Warming over land increases the severity and frequency of droughts around much of the world.[47][48]:1057 In some tropical and subtropical regions of the world, there will probably be less rain due to global warming. This will make them more prone to drought. Droughts are set to worsen in many regions of the world. These include Central America, the Amazon and south-western South America. They also include West and Southern Africa. The Mediterranean and south-western Australia are also some of these regions.[48]:1157

Higher temperatures increase evaporation. This dries the soil and increases plant stress. Agriculture suffers as a result. This means even regions where overall rainfall is expected to remain relatively stable will experience these impacts.[48]:1157 These regions include central and northern Europe. Without climate change mitigation, around one third of land areas are likely to experience moderate or more severe drought by 2100.[48]:1157 Due to global warming droughts are more frequent and intense than in the past.[49]

Several impacts make their impacts worse. These are increased water demand, population growth and urban expansion in many areas.[50] Land restoration can help reduce the impact of droughts. One example of this is agroforestry.[51]

Erosion and human activities

See also: Aeolian processes

Human activity can directly trigger exacerbating factors such as over-farming, excessive irrigation,[52] deforestation, and erosion adversely impact the ability of the land to capture and hold water.[53] In arid climates, the main source of erosion is wind.[54] Erosion can be the result of material movement by the wind. The wind can cause small particles to be lifted and therefore moved to another region (deflation). Suspended particles within the wind may impact on solid objects causing erosion by abrasion (ecological succession). Wind erosion generally occurs in areas with little or no vegetation, often in areas where there is insufficient rainfall to support vegetation.[55]

Impacts



Global drought total economic loss risk


Pair of dead oryx in Namibia during the 2018–19 Southern Africa drought.


After years of drought and dust storms the town of Farina in South Australia was abandoned.

Drought is one of the most complex and major natural hazards, and it has devastating impacts on the environment, economy, water resources, agriculture, and society worldwide.[22]

One can divide the impacts of droughts and water shortages into three groups: environmental, economic and social (including health).

Environmental and economic impacts


Western red cedar dying from drought, USA, 2018

Environmental effects of droughts include: lower surface and subterranean water-levels, lower flow-levels (with a decrease below the minimum leading to direct danger for amphibian life), increased pollution of surface water, the drying out of wetlands, more and larger wildfires, higher deflation intensity, loss of biodiversity, worse health of trees and the appearance of pests and dendroid diseases.[56][6]

Economic losses as a result of droughts include lower agricultural, forests, game and fishing output, higher food-production costs, lower energy-production levels in hydro plants, losses caused by depleted water tourism and transport revenue, problems with water supply for the energy sector and for technological processes in metallurgy, mining, the chemical, paper, wood, foodstuff industries etc., disruption of water supplies for municipal economies.

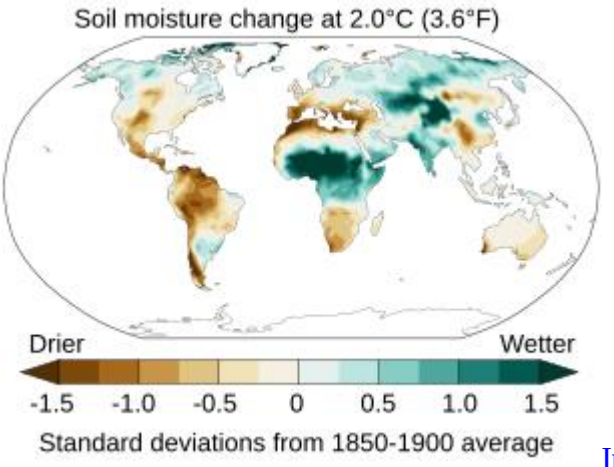
Further examples of common environmental and economic consequences of drought include:

- Alteration of diversity of plant communities, which can have an impact on net primary production and other ecosystem services.[57]

- Wildfires, such as Australian bushfires and wildfires in the United States, become more common during times of drought and may cause human deaths.[58]
- Dust Bowls, themselves a sign of erosion, which further erode the landscape
- Dust storms, when drought hits an area suffering from desertification and erosion
- Habitat damage, affecting both terrestrial and aquatic wildlife[59]
- Snake migration, which results in snake-bites[60]
- Reduced electricity production due to reduced water-flow through hydroelectric dams[61]
- Shortages of water for industrial users[62][63]

Agricultural impacts

Further information: Effects of climate change on agriculture



Impacts of climate change on soil moisture at 2 °C of global warming. A reduction of one standard deviation means that average soil moisture will approximate the ninth driest year between 1850 and 1900.

Droughts can cause land degradation and loss of soil moisture, resulting in the destruction of cropland productivity.[64] This can result in diminished crop growth or yield productions and carrying capacity for livestock. Drought in combination with high levels of grazing pressure can function as the tipping point for an ecosystem, causing woody encroachment.[65]

Water stress affects plant development and quality in a variety of ways: firstly drought can cause poor germination and impaired seedling development.[66] At the same time plant growth relies on cellular division, cell enlargement, and differentiation. Drought stress impairs mitosis and cell elongation via loss of turgor pressure which results in poor growth.[67] Development of leaves is also dependent upon turgor pressure, concentration of nutrients, and carbon assimilates[*clarification needed*] all of which are reduced by drought conditions, thus drought stress lead to a decrease in leaf size and number.[67] Plant height, biomass, leaf size and stem girth has been shown to decrease in maize under water limiting conditions.[67] Crop yield is also negatively effected by drought stress, the reduction in crop yield results from a decrease in photosynthetic rate, changes in leaf development, and altered allocation of resources all due to drought stress.[67] Crop plants exposed to drought stress suffer from reductions in leaf water potential and transpiration rate. Water-use efficiency increases in crops such as wheat while decreasing in others, such as potatoes.[68][69][67]

Plants need water for the uptake of nutrients from the soil, and for the transport of nutrients throughout the plant: drought conditions limit these functions leading to stunted growth. Drought stress also causes a decrease in photosynthetic activity in plants due to the reduction of photosynthetic tissues, stomatal closure, and reduced performance of photosynthetic machinery. This reduction in photosynthetic activity contributes to the reduction in plant growth and yields.[67] Another factor influencing reduced plant growth and yields include the allocation of resources; following drought stress plants will allocate more resources to roots to aid in water uptake increasing root growth and reducing the growth of other plant parts while decreasing yields.[67]

Social and health impacts

The most negative impacts of drought for humans include crop failure, food crisis, famine, malnutrition, and poverty, which lead to loss of life and mass migration of people.[22]

There are negative effects on the health of people who are directly exposed to this phenomenon (excessive heat waves). Droughts can also cause limitations of water supplies, increased water pollution levels, high food-costs, stress caused by failed harvests, water scarcity, etc. Reduced water quality can occur because lower water-flows reduce dilution of pollutants and increase contamination of remaining water sources.[70][71]

This explains why droughts and water scarcity operate as a factor which increases the gap between developed and developing countries.[72]

Effects vary according to vulnerability. For example, subsistence farmers are more likely to migrate during drought because they do not have alternative food-sources. Areas with populations that depend on water sources as a major food-source are more vulnerable to famine.


People displaced by a drought in Somalia arriving at a camp in Dolo Ado, Ethiopia, 2011

Further examples of social and health consequences include:

- Water scarcity, crop failure, famine[73] and hunger – drought provides too little water to support food crops; malnutrition, dehydration and related diseases
- Mass migration, resulting in internal displacement and international refugees
- Social unrest
- War over natural resources, including water and food
- Cyanotoxin accumulation within food chains and water supply (some of which are among the most potent toxins known to science) can cause cancer with low exposure over the long term.[74] High levels of microcystin appeared in San Francisco Bay Area salt-water shellfish and fresh-water supplies throughout the state of California in 2016.

Loss of fertile soils

See also: Aeolian processes

Wind erosion is much more severe in arid areas and during times of drought. For example, in the Great Plains, it is estimated that soil loss due to wind erosion can be as much as 6100 times greater in drought years than in wet years.[75]

Loess is a homogeneous, typically nonstratified, porous, friable, slightly coherent, often calcareous, fine-grained, silty, pale yellow or buff, windblown (Aeolian) sediment.[76] It generally occurs as a widespread blanket deposit that covers areas of hundreds of square kilometers and tens of meters thick. Loess often stands in either steep or vertical faces.[77] Loess tends to develop into highly rich soils. Under appropriate climatic conditions, areas with loess are among the most agriculturally productive in the world.[78] Loess deposits are geologically unstable by nature, and will erode very readily. Therefore, windbreaks (such as big trees and bushes) are often planted by farmers to reduce the wind erosion of loess.[54]

Protection, mitigation and relief

See also: Water scarcity, Water supply, and Water security


Water distribution on Marshall Islands during El Niño.

Agriculturally, people can effectively mitigate much of the impact of drought through irrigation and crop rotation. Failure to develop adequate drought mitigation strategies carries a grave human cost in the modern era, exacerbated by ever-increasing population densities.

Strategies for drought protection or mitigation include:

- Dams – many dams and their associated reservoirs supply additional water in times of drought.[99]
- Cloud seeding – a form of intentional weather modification to induce rainfall.[100] This remains a hotly debated topic, as the United States National Research Council released a report in 2004 stating that to date, there is still no convincing scientific proof of the efficacy of intentional weather modification.[101]
- Land use – Carefully planned crop rotation can help to minimize erosion and allow farmers to plant less water-dependent crops in drier years.
- Transvasement – Building canals or redirecting rivers as massive attempts at irrigation in drought-prone areas.

When water is scarce due to droughts, there are a range of options for people to access other sources of water, such as wastewater reuse, rainwater harvesting and stormwater recovery, or seawater desalination.

Drought has many impacts, including:

- Environmental

  Droughts can harm plant, animal, and fish species, and alter the balance of natural systems. Droughts can also lead to:

  - Wildfires: Dry grasses and trees become more flammable during droughts, increasing the risk of wildfires.
  - Invasive species: Native plants may die back, allowing invasive species to take over and disrupt the ecosystem.
  - Desertification: Long-term droughts can lead to desertification.
- Agricultural

  Droughts can limit growing seasons, which can lead to:

  - Crop failure: Crops and forage may fail completely.

    - Rising food prices: Low crop yields can lead to rising food prices and shortages.
    - Livestock sell-offs: Livestock may be sold off.
- Health

  Droughts can lead to:

    - Illness and disease: Droughts can increase the incidence of illness and disease.
    - Respiratory illnesses: Particulate matter from wildfires and dust storms can irritate the lungs and bronchial passages.
    - Mental health issues: Droughts can lead to mental health issues related to job and economic losses.
- Displacement
  Droughts can lead to people being displaced from their homes. People may have to travel further to find clean water, which can fall to women and children

Droughts can have many hazards in India, including:

- Agriculture

  Droughts can cause agricultural productivity to drop by up to 40% in rainfed areas. In parts of India, such as southeastern Maharashtra, northern Karnataka, Andhra Pradesh, Odisha, Gujarat, Telangana, and Rajasthan, droughts can lead to poor yields due to water shortages.
- Water availability

  Droughts can impact water availability for people and cattle.
- Employment

  Rural populations whose livelihoods are dependent on agriculture lose employment opportunities.
- Malnutrition

  Insufficient nutrition causes malnutrition, making people more susceptible to infection and disease.
- Economic impact

  Droughts can lead to a reduction in per capita income and a loss of employment days.
- Wildfires
  During drought conditions, fuels for wildfire, such as grasses and trees, can dry out and become more flammable.

Some of the regions in India that are prone to drought include Punjab, Haryana, Uttar Pradesh, and Tamil Nadu.

To stay aware of droughts, you can keep an eye out for government alerts. You can also consider taking out life insurance or a drought-specific policy.

There are several measures that can be taken to control drought, including:

- Water conservation

  Using water wisely can help ensure that more water is available during a drought.

- Irrigation

  Providing sufficient water to plants and trees is an efficient way to control drought.

- Rainwater harvesting

  Installing rainwater harvesting systems in buildings and community-level structures can help increase groundwater levels.

- Crop engineering

  Modifying the genetics of crops can help them be more resistant to drought and increase their yield.

- Drought-resistant crops

  Growing crops that require less water, such as millets, pulses, and oilseeds, can help mitigate drought.

- Watershed management and afforestation

  Planting trees can help hold moisture in the soil and reduce surface runoff.

- Desalination

  Brackish and salt water can be used to supplement water supplies in areas affected by drought.

- Reuse of treated wastewater

  Treating wastewater to irrigation standards can help reduce freshwater pressure.

- Water budgeting and monitoring

  Preparing basin-wise water budgets can help plan optimal water usage.

- Drought forecasting
  An early warning of drought can help reduce the effects of droughts

Lecture No 8

Natural Hazards and Disasters

A. Floods

Flood is the rise of water levels which are abnormally high and inundated neighbouring areas. This could be because of heavy precipitation in the river basin, dam failure, rapid snow melts, cloud bursts, glacial lake outbursts, storm surges etc.

Floods are of three types

a) River Floods

b) Flash Floods

c) Coastal Floods

Causes of flood

The causes of flood are:

(i) Heavy rainfall: Heavyrain in the catchment area of a river causes water rush downstream resulting in flooding in nearby areas.

(ii) Sediment deposition: Siltation in river course causes reduction in capacity to carry water. As a result, the heavy rain waters over flows the river banks.

(iii) Deforestation: Vegetation hampers the quick flow of water and forces it to percolate in the ground. Barren land surface is subject to soil erosion. Obstruction-free surroundings in the catchment area of a channel cause flooding during the rainy season.

(iv) Cyclones: It generates sea waves of abnormal height (storm surges) and causes spreads of water in the adjoining coastal areas.

(v) Interference in drainage system: Drainage congestion caused by badly planned construction of bridges, roads, railway tracks, canals etc. hampers the flow of water and the result is flooding.

(vi) Natural Change in the course of the river: As river changes, it course in inundated new areas

(vii) Tsunami: Large coastal areas are flooded by rising sea water because of underwater earthquakes.

(viii) Dam Failure: due to earthquake or human induced factors dams are damaged and broken causing flooding downstream.

DROUGHT

Drought has economic, environmental and social impacts. In India, around 68% of the countryis prone to drought in

varying degrees. Area receiving rainfall between 75 cm to 112.5cm isconsidered drought prone constituting 35% of total area. With rainfall less than 75 cm ischronically drought prone and it covers 33% of the total area of the country.

According to meteorologists the rainfall deficiency during a long period over a large area iscalled a drought. IMD defines Drought as a situation occurring in any area when mean annualrainfall is less than 75% of the normal rainfall. Some times in Hindi language famine Akal andAnavrishty are also used for drought. Drought can also occur when ground water level is notwithin reach of agricultural communities. The government also declares drought, if more than 50 percent crop loss happens in an area due to meteorological conditions.

Causes of Drought

Major cause of drought is scarcity of rain. But humans have interfered in the natural processesby their activities. People have filled up the natural resources like ponds and lakes. They haveremoved the vegetation cover. Vegetation cover impedes the flow of rainwater and forces it topercolate. Over utilisation of ground water resources through tube wells is drying groundaquifers. Inadequate rainfall changes in local landscape, meteorological, hydrological, agricultural factors cause drought.

Impact of Drought

Droughts cause scarcity of food and water. People die of hunger, malnutrition and epidemics. Crops fail due to scarcity of water. Cattle are either small nourished or die due to shortage of fodder and water. Farmers are deprived of their employment. People are forced to migrate out of their villages in search of livelihood.

B. CYCLONES

Cyclones are the centre of low atmospheric pressure. The air pressure increases from the centre to the outer areas. Consequently winds blow from outside to the centre. In cyclones, winds blow in an anticlockwise direction in the northern hemisphere and clockwise in the southern hemisphere. On the basis of their location and physical properties cyclones are of two types; temperate cyclones and tropical cyclones.

Cyclone is a violent circular storm, winds. It is associated with torrential rain, high speed winds and sea surges (water raises). Cyclones play an important role in the general circulation of the atmosphere.

Occurrence and movement of Cyclones

Cyclones have seasonal cyclic patterns. It originates over the sea surface and dissipates as they reach land. In India, most of the cyclone occurrence is concentrated in the post monsoon season, i.e. from October to December or in the pre-monsoon season from April to May. The lifespan of a cyclone can range between 7 to 14 days from its development to landfall. The cyclone moves forward from east to west in Bay of Bengal with an initial.

Speed of 15 to 30 km per hour. Later on the speed reaches to over 90 KM/hour and in extra cases it goes beyond 250 KM/hour. High sea waves are generated in the sea by speedy cyclonic winds. They strike the coastal areas like high walls of water and flood the areas upto 10-15 km from the coast. The impact can be felt upto 50 kms inland. The cyclone that struck

Odisha originated near Andaman & Nicobar Islands and reached Orissa on 29-10-1999 after many days. The movement of a cyclone in a direction is like the movement of a spinning top.

Impacts of cyclones and floods

(i) Damage to build up structures: The violent winds of a cyclone destroy whatever comes in their way from; thatched huts to the palaces, houses, forts, lines of electricity and communication.

(ii) Natural Habitat: High speed winds cause damage to vegetation. Trees are uprooted. Salt water intrudes in coastal areas leaving the soil infertile.

(iii) Floods: Torrential rain causes floods. Floods wreak havoc all around.

C. EARTHQUAKE

Earthquakes can occur at any time. Its impact is very sudden. There are no warning signs of earthquakes. Earthquake is sudden shaking or trembling of the earth surface caused by passage of seismic waves. Most earthquakes are minor tremors. Larger earthquakes usually begin with slight tremors but suddenly they turn into violent shocks and after that they diminish. Tremors or shocks are felt for a few seconds only.

In spite of extensive research, it is not possible to forecast or predict earthquakes. INSAR (Interferometry Synthetic Aperture Radars) are used in certain earthquake prone areas but they can only predict just a few seconds before an earthquake strikes.

Impact of Earthquakes

(i) **Damage of Property:** When an earthquake occurs, buildings from cottages to palaces and skyscrapers are subject to damage. Underground pipelines and railway lines are damaged or broken. Dams on river collapse, resultant floods cause havoc. The earthquake in 1967 in Koyna damaged the Dam.

(ii) **Human Loss -** Normally the duration of earthquake tremors is only a few seconds, but thousands of people may die in this short period. The Bihar earthquake of 1934 killed 10,000 people and the Kangra earthquake of 1905 caused 20,000 people dead. Numerous people lost their shelter and many became orphans. The earthquake that occurred in Gujarat on 26 January 2001 was devastating. More than 25,000 people died due to the impact of this earthquake. The destruction of property was tremendous.

(iii) **Changes in River Courses:** Sometimes river channels are blocked or their courses are changed due to the impact of an earthquake.

(iv) **Tsunamis:** are caused by underwater earthquakes which trigger very high sea waves over the coastal areas. It wreaks havoc on settlements of coastal areas. It sinks large ships. Tsunami that occurred on 26-12-2004 near the coast of Sumatra (Indonesia) damaged property worth billions of rupees. More than two lakh people lost their lives in Southeast Asia, India and Sri Lanka.

(v) **Fountains of Mud:** Due to the intense impact of the earthquake, water and mud appear on the surface and take the form of fountains. It leads to spreading sand on agricultural land and making it infertile.

(vi) **Cracks and Fissures:** Sometimes cracks and fissures develop in roads, railway tracks, and fields.

(vii) Landslides and avalanches are triggered by earthquakes

Floods

Reservoirs: By constructing reservoirs in the course of rivers could storeextra water at the time of flood.

(ii) **Embankments:** By building flood protection embankments, flood water can be controlled from overflowing the banks and spreading in nearby areas.

(iii) **Afforestation:** The fury of flood could be minimised by planting trees in catchment areas of rivers.

Some important precautions:

- Avoid building houses in flood prone areas unless you elevate and reinforce your home.
- Elevate the furnace, water heater, and electric panel if susceptible to flooding.
- Install "Check Valves" in the sewer to prevent backing floodwater.
- Contact community or government officials to find out if they are planning to construct levees or floodwalls.
- Seal the walls in your basement with waterproofing.
- Listen to the radio or television for information.
- Be aware of flash flooding. If there is any possibility of flood, move immediately to higher ground.
- Be aware of streams, drainage channels and other areas known to flood suddenly.
- Secure your home. If you have time, shift outdoor furniture and essential items to the upper floor.
- Disconnect electrical appliances. Do not touch electrical equipment if you are wet or standing in water.
- Do not walk through running water.
- Do not drive into flooded areas. If floodwaters rise around your car, abandon the car and move to higher ground.

Droughts(i) Suitable farming methods for arid areas: Production of coarse and hardy cereals; conservation of soil moisture by deep ploughing, storing water behind small dams, collecting water in ponds and tanks and use of sprinklers for irrigation.

(ii) Sowing drought resistant crops: By sowing drought resistant crops like cotton, moong, pearl millet, wheat etc, the impact ofdrought could be minimised to a certain extent.

(ii) Rain water harvesting: Collection of each and every drop of rain couldhelp in coping with the drought.

(iii) Making high bunds around the fields, planting trees on the bunds of fields, the use of rainwater can be maximised.

(iv) Irrigation canal construction with mortar and bricks

Small quantity of water can irrigate a comparatively larger area by using adrip irrigation method

Landslide

- Database of vulnerable sites and location
- Awareness and development of both indigenous and modern methods
- Detailed landslide hazard zonation mapping
- Afforestation : Trees and brushes help in binding the soil particles.
- Ban on quarrying of stones and mining of minerals in fragile areas
- Permanent crops like orchards of fruits should replace the seasonal or annual crops.
- Retaining walls can be built on mountain slopes.
- Ban construction activities in landslide prone areas/spots.

Earthquake

- Analysis of seismic risk and its zonation;
- Design and construction of earthquake resistant structures.
- Retrofitting of existing structures strengthening them.
- Earthquake insurance for houses to reduce the economic impact on individuals.
- Installation of seismological observations for monitoring seismic activity.

Cyclones □ Check the house; secure loose tiles and carry out repairs of doors and windows

- Remove dead branches or dying trees close to houses;

- Fixing of removable objects such as lumber piles, loose tin sheets, loose bricks, garbage cans, sign-boards etc.
- Keep some wooden boards ready so that glass windows can be boarded ifneeded
- Keep a cyclone lantern filled with kerosene, battery operated torches andenough dry cells
- Demolish condemned buildings
- Keep some extra batteries for transistors
- Keep some dry non-perishable food always ready for use in emergency
- Listen to the radio for weather warnings.
- Keep monitoring the warnings. This will help you prepare for a cyclone emer-gency.
- Pass the information to others.
- Believe in the official information
- When a cyclone alert is on for your area, stay alert to the warnings.

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